



DEPARTMENT OF EDUCATION
ANDAMAN & NICOBAR ADMINISTRATION



FUN WITH COMPUTERS

GUIDE TO BASIC COMPUTING



Cyber Safety!

Think Before You Click
Cyber Safety Matters!

Cybercrime helpline number

1930

9531856063

6

GRADE



State Institute of Education
Shiksha Sadan, Sri Vijaya Puram
Andaman & Nicobar Islands

in collaboration with
State Project Office, Samagra Shiksha
UT Mission Authority, A & N Islands



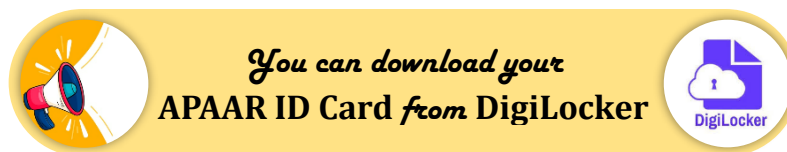
APAAR - One Nation, One Student ID

12 Digit ID | Track education journey



The Automated Permanent Academic Account Registry (APAAR) is a unique identification system introduced by the Government of India to streamline and manage students' academic records nationwide.

As part of the 'One Nation, One Student ID' initiative under the National Education Policy (NEP) 2020, APAAR assigns each student a lifelong 12-digit ID, facilitating seamless tracking of their educational journey from early schooling through higher education.



FUN WITH COMPUTERS

GUIDE TO BASIC COMPUTING

CLASS 6

State Institute of Education

Shiksha Sadan, Sri Vijaya Puram, Andaman & Nicobar Islands
in collaboration with

State Project Office, Samagra Shiksha

UT Mission Authority, A & N Islands



All rights reserved
Department of Education
Andaman & Nicobar Administration

No part of this book may be reproduced, distributed or transmitted in any form or by any means including photocopying, recording or other electronic or mechanical methods without the prior permission of the Department of Education, Andaman & Nicobar Administration, except as permitted by Indian Copyright Act 1957.

विक्रम सिंह, दानिक्स
Vikram Singh, DANICS
शिक्षा निदेशक
Director (Education)



शिक्षा निदेशालय
Directorate of Education
अंडमान एवं निकोबार प्रशासन
Andaman & Nicobar Administration
श्री विजयपुरम / Sri Vijaya Puram

MESSAGE

In alignment with the National Education Policy 2020's vision of integrating skill education, the State Institute of Education, Sri Vijaya Puram, in collaboration with Samagra Shiksha, UT Mission Authority, A & N Islands has developed comprehensive Computer Books for Grades 6-8. These books seamlessly blend theoretical and practical computer knowledge, addressing the evolving digital landscape.

In today's rapidly evolving digital world, it is crucial to empower the students to navigable real-life challenges such as cyber fraud, phishing, digital arrests, and money scams, offering locally relevant solutions and preventive measures.

To ensure the highest quality, the books were reviewed by a team of experts from the Department of Computer Science, JNRM. The content is presented in simple, student-friendly language and enhanced with vibrant visuals, including colorful pictures, images, and graphic content, to foster engagement and comprehension.

I believe these Computer Books will serve as a vital resource, empowering Upper Primary Level students with essential computer skills and cultivating their logical thinking abilities, ultimately contributing to a more technologically proficient generation.

Vikram Singh, DANICS
Director (Education)

राज्य शिक्षा संस्थान
शिक्षा सदन, श्री विजयपुरम
अंडमान और निकोबार द्वीप समूह



State Institute of Education
Shiksha Sadan, Sri Vijaya Puram
Andaman & Nicobar Islands

FOREWORD

The Department of Education, Andaman and Nicobar Islands, with the support of Samagra Shiksha, UT Mission Authority, has made significant strides in enhancing the educational infrastructure across the islands. The ICT Lab facilities have been successfully established in 146 Government Middle, Secondary and Senior Secondary schools. These ICT labs have been equipped with proper lab facilities, computer Systems, internet access, and an ICT Instructor.

Despite these advancements, students at the Middle and Secondary levels have faced challenges in fully utilizing these facilities due to the lack of a structured curriculum and syllabus. To address this, the State Institute of Education, in collaboration with a team of Computer Teachers and subject experts, developed a Teaching Manual for students at the Middle Level (Class VI – VIII) in 2023. Building on this foundation, the SIE has now developed Computer Books for students of Class VI, VII, and VIII, which will be distributed to schools.

The CBSE has also provided an outline for the curriculum of IT as a Skill Subject at the Middle Level. The course content is designed to empower students to work independently on computers and develop applications for small-scale computations and record-keeping. This curriculum aims to foster Computational and Thinking Skills, Problem Solving Skills, Decision Making, and logical and analytical abilities among students.

This course will not only impart fundamental computer knowledge but also instil confidence in students to develop small applications. This initiative will significantly contribute to the holistic development of our students, preparing them for the challenges of the digital age. I am excited to embark on this journey of integrating ICT-based skills into our education system and look forward to the positive impact it will have on our students' learning experiences.



BOOK DEVELOPMENT TEAM

PATRON

- ★ Mr. Vikram Singh, DANICS, Director (Education) / State Project Director, Samagra Shiksha
- ★ Mr. S. Suresh Kumar, State Project Officer, Samagra Shiksha
- ★ Ms. Sangeeta Chand, Principal, State Institute of Education

ADVISORY TEAM

- ★ Mr. T. Samson, HoD (ICT)/Project Officer, Samagra Shiksha
- ★ Mr. M. Panneer Selvam, Project Officer, Samagra Shiksha
- ★ Mr. K.S. Pradeep, Assistant Project Officer (IT), Samagra Shiksha
- ★ Mr. S.L. Vinjit, Faculty (ICT), SIE

CONTENT DEVELOPMENT TEAM

- ★ Mr. K Subramaniam, Post Graduate Teacher (Computer Science)
- ★ Ms. Lakshmi Prasanna, Post Graduate Teacher (Computer Science)
- ★ Ms. C P Pradika, Craft Instructor (Computer Applications)
- ★ Mr. D David, Vocational Teacher (Information Technology)
- ★ Ms. G Jayalaxmi, Vocational Teacher (Information Technology)
- ★ Ms. Sherin Mahiba P, Vocational Teacher (Information Technology)
- ★ Ms. R Lakshmi, Vocational Teacher (Information Technology)

REVIEW TEAM (JNRM)

- ★ Dr. Vinod N C, HoD, Computer Science
- ★ Mr. Rajeev Kumar Tiwari, Assistant Professor
- ★ Dr. C Parthiban, Assistant Professor
- ★ Dr. T Hemalatha, Assistant Professor
- ★ Ms. Shaniba P M, Guest Faculty
- ★ Ms. A M Eadel Queen, Guest Faculty
- ★ Ms. G Vanitha, Guest Faculty
- ★ Mr. M Kathirvel, Guest Faculty
- ★ Ms. T P Sneha, Guest Faculty
- ★ Ms. C P Priyanka, Guest Faculty

COMPUTER DESIGNING

- ★ Mr. K.S. Pradeep, Assistant Project Officer (IT), Samagra Shiksha
- ★ Ms. Geeta Devi, CI (Computer Applications), SIE

MEMBER COORDINATORS

- ★ Mr. T. Samson, HoD (ICT)/Project Officer, Samagra Shiksha- Coordinator
- ★ Mr. S.L. Vinjit, Faculty (ICT), SIE - Asst. Coordinator

Table of

Contents

1	Fundamentals of Computers-1 1
2	Fundamentals of Computers-2..... 15
3	Introduction to Windows..... 29
4	MS Paint & Paint 3D 47
5	Introduction to Word Processing (MS Word) 63
6	Computer Networks 81
7	Cyber Ethics..... 91

INTRODUCTION TO COMPUTERS-1

Learning objectives

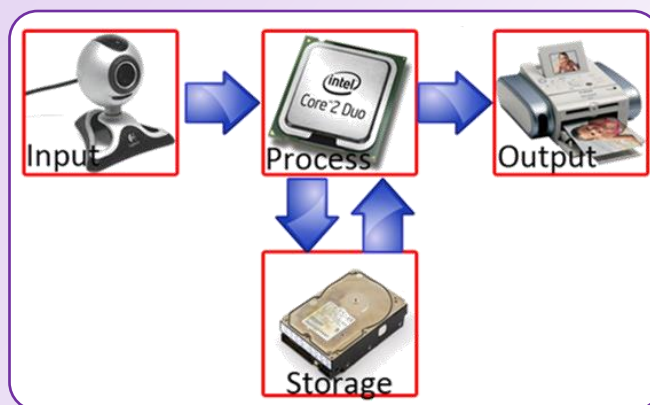
In this chapter, you will be able to understand:

- ★ Introduction to Computers
- ★ Characteristics of a Computer
- ★ Brief history of Computers
- ★ Components of a Computer
- ★ Input/ Output Devices
- ★ Software and its types - Application and System Software.



INTRODUCTION TO COMPUTERS

Being a modern-day kid you must have used, seen, or read about computers. This is because they are an integral part of our everyday existence. Be it school, banks, shops, railway stations, hospital or your own home, computers are present everywhere, making our work easier and faster for us. As they are such integral parts of our lives, we must know what they are and how they function. Let us start with defining the term computer formally.



Computer is an electronic device that accepts data from the user (Input), Process it and produces meaningful information to the user (Output). Storage is a mechanism that enables a computer to retain data, either temporarily or permanently.

CHARACTERISTICS OF A COMPUTER



Speed

A computer works with much higher speed and accuracy compared to humans while performing mathematical calculations. The time taken by computers for their operations is microseconds and nanoseconds.



Accuracy

Computers perform calculations with 100% accuracy. Errors may occur due to data inconsistency or inaccuracy.



Diligence

A computer can perform millions of tasks or calculations consistently without tiring.

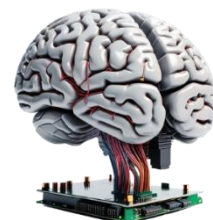


Versatility (Flexibility)

Versatility refers to the capability of a computer to perform different kinds of works with same accuracy and efficiency.

DO YOU KNOW???

A CPU is called as the **BRAIN OF COMPUTER**





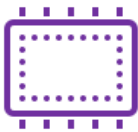
Reliability

The results obtained by the computer are very reliable. But this is true only when the data given to the computer is correct.



Automation

Computer can also be used to automate the routine tasks such as sending an email, scanning for viruses and other maintenance tasks. Besides computer can be programmed to perform many complex task too.



Memory

Memory is the ability to store and retrieve information when people need it. A computer has Primary and Secondary storage such as RAM, ROM, HARD DISK, CDs, Pen drives etc.

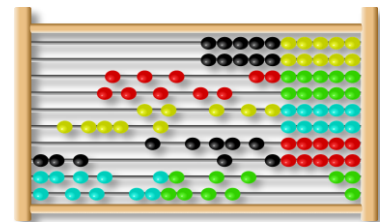
BRIEF HISTORY OF COMPUTERS

Since the evolution of humans, devices have been used for calculations for thousands of years. Then in 1822, the father of computers, **Charles Babbage** began developing what would be the first mechanical computer.



Abacus

Abacus is a wooden rack with metal rods with beads attached to them. The abacus operator moves the beads according to certain guidelines to complete arithmetic computations.



DO YOU KNOW???

LADY ADA LOVELACE is the first Computer Programmer.



First Generation Computer (1940-1956)



Second Generation Computer (1957-1963)



Third Generation Computer (1964-1971)



Fourth Generation Computer (1972 onward)

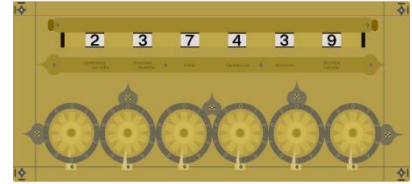


Fifth Generation Computer (Present)



Pascaline

Pascaline was invented in 1642 by Blaise Pascal, a French mathematician and philosopher. It is thought to be the first mechanical and automated calculator. It was a wooden box with gears and wheels inside.



Difference Engine

In the early 1820s, Charles Babbage created the Difference Engine. It was a mechanical calculator designed to perform calculations automatically. It was a steam-powered calculating machine used to solve numerical tables such as logarithmic tables.



Analytical Engine

Charles Babbage created another calculating machine, the Analytical Engine, in 1830. It was a mechanical computer that took input from punch cards. It was capable of solving any mathematical problem and storing data in an indefinite memory.



DO YOU KNOW???

CHARLES BABBAGE is known as the
“Father of Computers”.



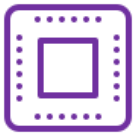
MAIN PARTS OF A COMPUTER

Monitor



It is the most important Output Device because people interact with this device most intensively than others. Information processed within the CPU is displayed through Monitor.

CPU (Central Processing Unit)



The CPU is the brain of a computer, containing all the circuitry needed to process input, store data, and output results. The processor contains three sections called the Arithmetic Logic Unit (ALU), the Control Unit and Memory Unit.



Keyboard



A computer keyboard is an input device that allows a person to enter letters, numbers, and other symbols (together, these are called characters) into a computer.

Mouse



It is an input device, also known as 'Pointing Device'. It is mainly used to select items on the computer screen.

INPUT DEVICES

The devices that are used to input data and programs in the computer are known as "Input Devices". Input unit accepts instructions and data from the user and converts these instructions and data in computer acceptable format, which are then sent to computer system for processing.



Keyboard

Keyboard is most common input device. The data and instructions are input by typing on the keyboard.



Mouse

It is a pointing device. The mouse is rolled over the mouse pad, which in turn controls the movement of the cursor on the screen.



Scanner

Scanners are used to enter information directly in to the computer memory.



Microphone

A microphone is an input device; it sends audio information to the computer.



Joystick

A joystick is an input device used to play virtual reality games.



OUTPUT DEVICES

An Output Device is a computer hardware component that is responsible for giving the data or information in a format which is easily understood by humans. These devices take electronic data from computer and convert it into various forms like text, visual, audio and hard copy(paper print).



Monitor

The monitor looks like a television screen. It is also **called Visual Display Unit (VDU)** and it is used to display information from the computer. There are colored as well as black and white monitors.



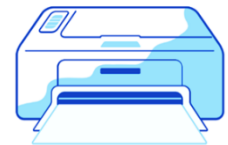
Speaker

A speaker is an output device that produces audio sound.



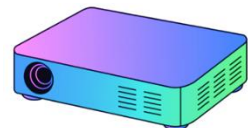
Printer

A printer is used for transferring data from the computer to the paper. There are colour printers as well as black and white printers.



Projector

A projector is an output device that allows you to display images, videos, or other content on a large screen or surface.



Headphones

With the help of headphones, we can listen to the audio privately and without disturbing anyone around.



DO YOU KNOW???

Words in **machine language** consists of **0's** and **1's**.



SOFTWARES

Software is a set of instructions or programs used to operate computers and execute specific tasks. Computer software consists of computer programs and libraries.



Android



iOS



Linux



Microsoft Windows

System software

System software is designed to run a computer's hardware and provides a platform for applications. System software includes Operating system (OS) like Windows, Linux, Unix, etc.

Application Software

An application can be self-contained, or it can be a group of programs that run the application for the user. Example: Word Processing Software, Spreadsheet Software, Presentation Software, Multimedia Software etc.



Let's Recapitulate



- ★ Computer is an electronic device that accepts Data from the user, Process it and Produces information to the user.
- ★ The major characteristics of the computer are Speed, Accuracy, Diligence, Versatility, Reliability, Consistency, Memory and Automation.
- ★ The devices that are used to input data and programs in the computer are known as "Input Devices".
- ★ The results of computer into human understandable form is called as Output devices.
- ★ Software is a set of instructions or programs used to operate computers and execute specific tasks.

Let's Practice



I. Fill in the blanks

1. _____ displays the output on the screen.
2. Data is entered into the computer through a process called _____.
3. _____ gives us the hard copy of the output.
4. We use _____ to type on a computer.
5. _____ is a pointing device.

II. Say true or false

1. Mouse have generally four buttons.
2. CPU stands for Central Processing Unit.
3. To print documents is the function of a printer.
4. The full form of ALU is All Logic Unit.
5. Speaker is used to listen to sound on the computer.

III. Multiple choice questions

1. Who is known as father of computer.
a) John Newton b) Charles Babbage
c) Melton d) None
2. The brain of any computer system is _____.
a) ALU b) Memory
c) CPU d) Control unit

3. Scanner is a _____.
 a) Output device b) Input device
 c) Input software d) None of these
4. _____ is the ability to store and retrieve information when people need it.
 a) Memory b) device
 c) software d) printer
5. Printer is a _____.
 a) control unit b) Input device
 c) Input software d) Output device

IV. Short answer type questions

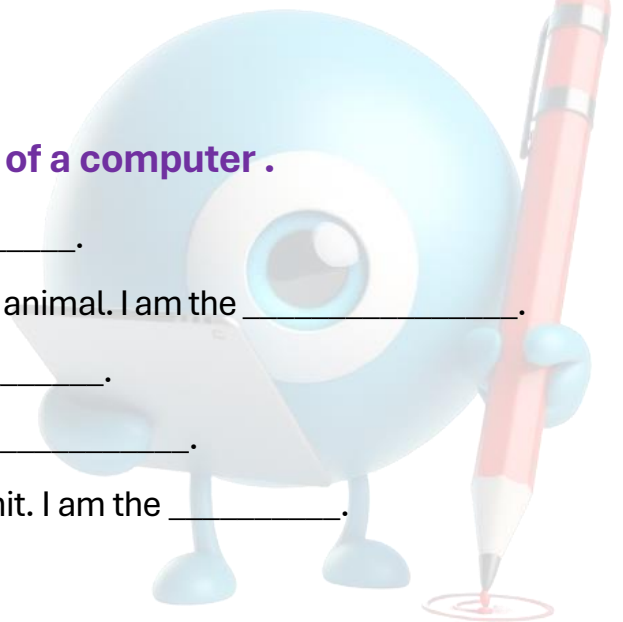
1. What are the components of computer?
2. What is an output device? Give two examples?
3. What is a projector?
4. What are the two main software types?
5. Name two common input devices?

Let's Do Hands on



I. Read the clues to name the parts of a computer .

1. I look like a TV. I am the _____.
2. My name is the same as that of an animal. I am the _____.
3. I have many keys. I am the _____.
4. I work like your brain. I am the _____.
5. I am also called Visual Display Unit. I am the _____.



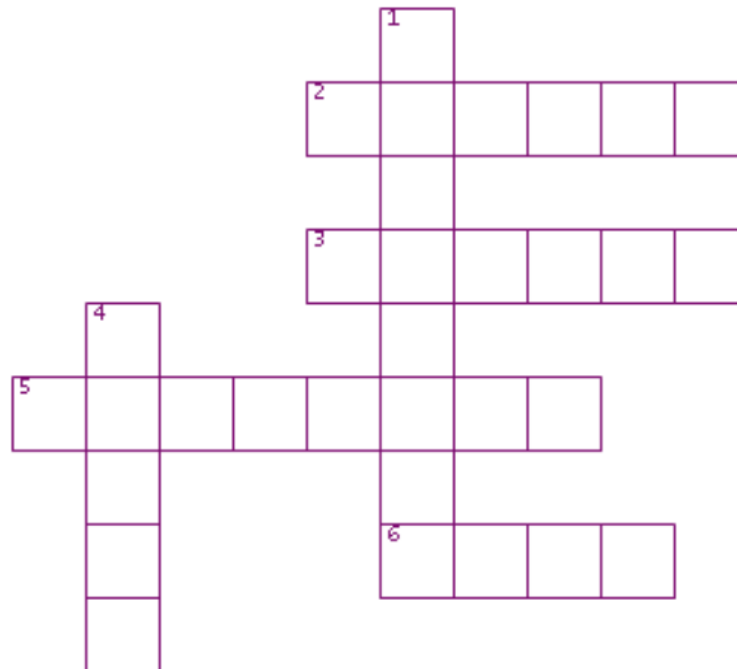
II. Solve the crossword puzzle using the hint(s) given below:

➡ Across

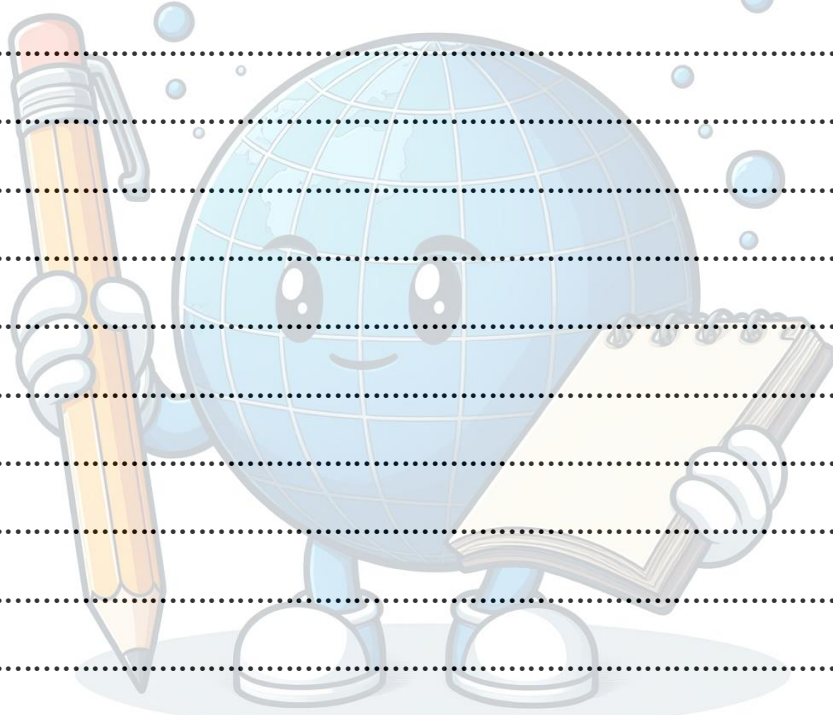
2. Ability to store & retrieve information
3. Wooden tool for arithmetic computations
5. Set of instructions or programs
6. It is entered through input devices

⬇ Down

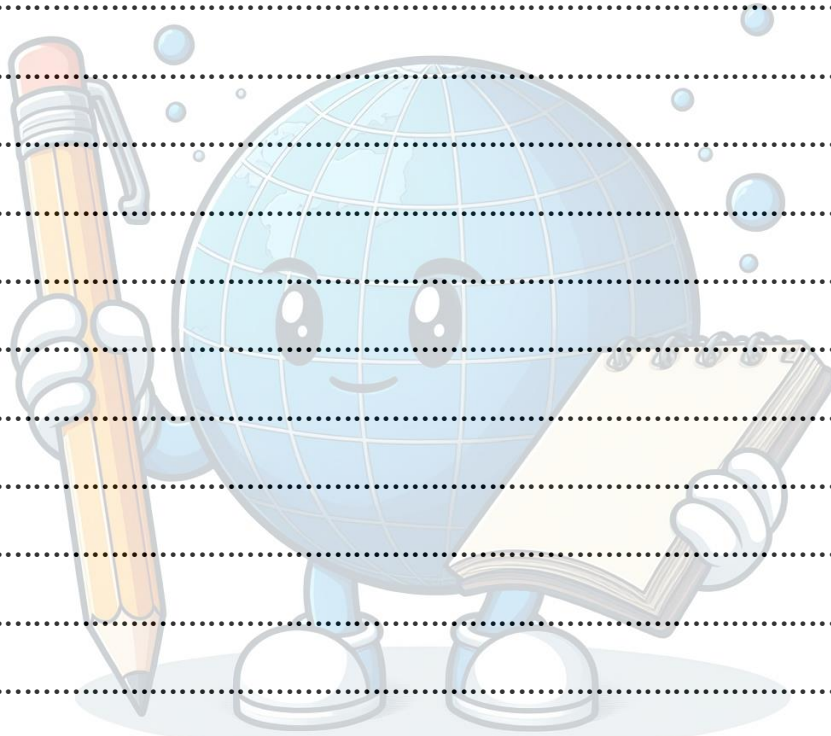
1. Input device for typing
4. Pointing input device



Space for Notes



Space for Notes



Answer Key



Let's Practice

I. Fill in the blanks

1. Monitor 2. Input 3. Printer 4. Keyboard 5. Mouse

II. Say true or false

1. False 2. True 3. True 4. False 5. True

III. Multiple choice questions

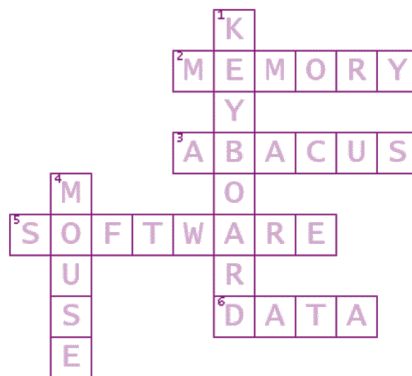
1. Charles Babbage 2. CPU 3. Input device 4. Memory 5. Output device

Let's Do Hands on

I. Read the clues

1. Monitor 2. Mouse 3. Keyboard 4. CPU 5. Monitor

II. Crossword puzzle



FUNDAMENTALS OF COMPUTERS-2

Learning objectives

In this chapter, you will be able to understand:

- ★ Switching On and Shutting Down of Computer
- ★ Different types of Hardware Devices
- ★ Functions of Keyboard and Mouse
- ★ Computer Memory & its types
- ★ Printer and its types



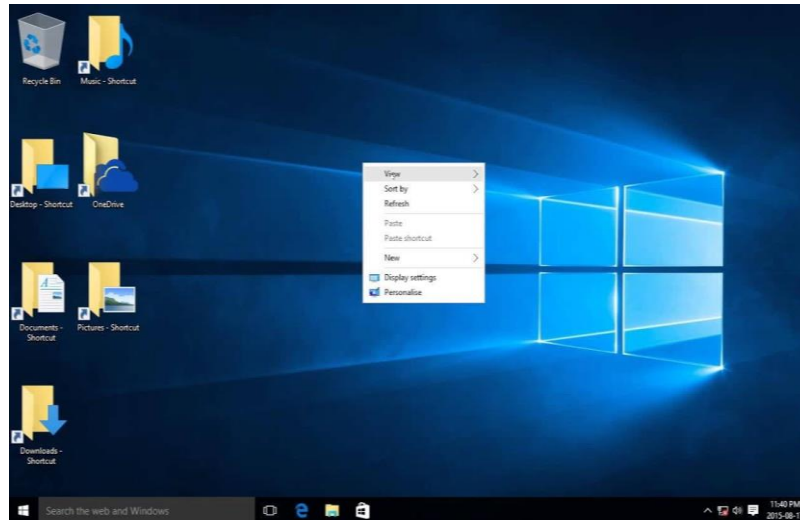
SWITCHING ON AND SHUTTING DOWN OF COMPUTER

To start a computer, follow these steps:

- Step 1 • Switch ON the Power Supply
- Step 2 • Switch ON the UPS
- Step 3 • Switch ON the CPU
- Step 4 • Switch ON the Monitor

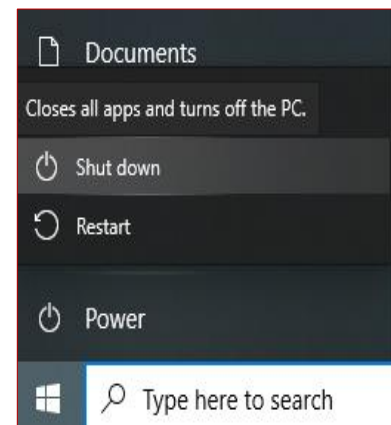


After switching on the monitor, wait for a while till you see a screen. This screen is called Desktop. The small pictures on the desktop are called icons. You can double click on these icons to start working on your computer.



To shut down the computer follow these steps:

- Step 1:** Move the mouse pointer to the start button on the desktop and left click on it.
- Step 2:** Click on power option and then click on shut down.
- Step 3:** A screen with a message 'Shutting down' will be seen. Wait for the CPU to turn off.
- Step 4:** Switch OFF the Monitor.
- Step 5:** Switch OFF the UPS.
- Step 6:** Switch OFF the Power supply.

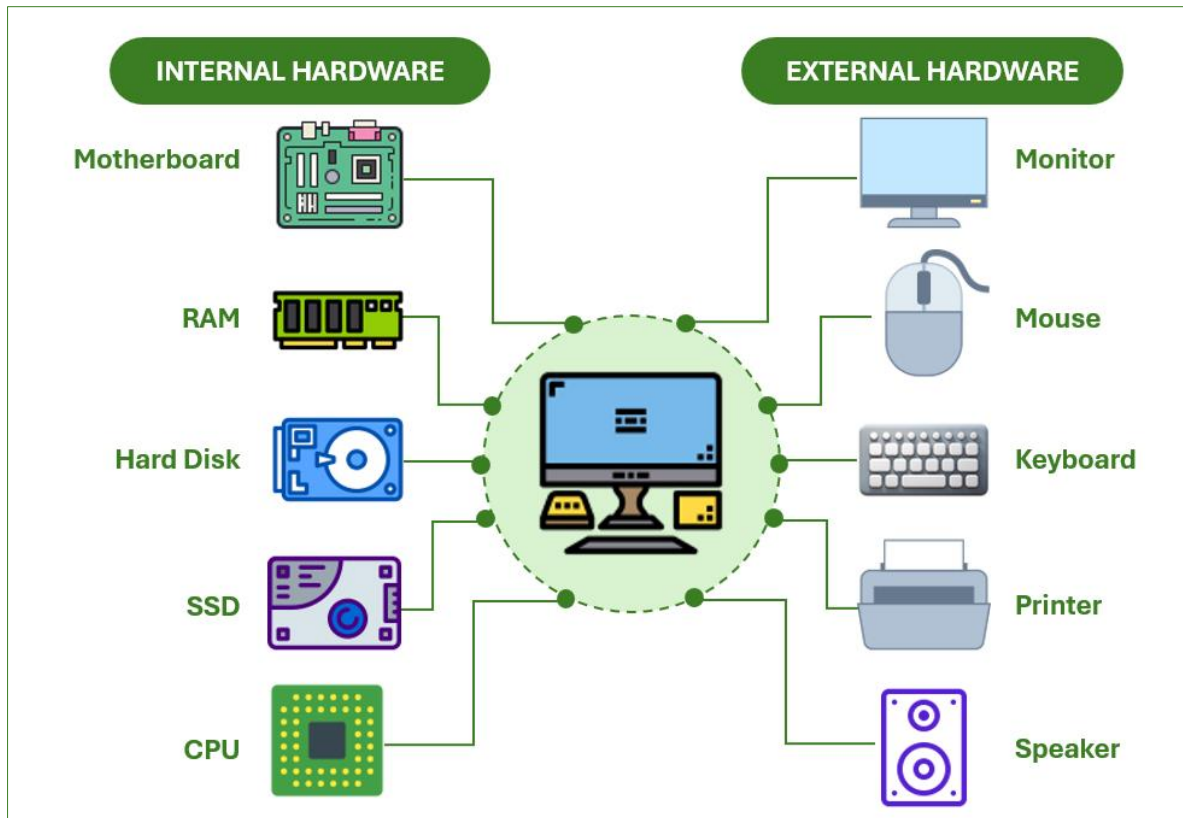


DIFFERENT TYPES OF HARDWARE DEVICES

Hardware refers to the external and internal devices and equipment that enable you to perform major functions such as input, output, storage, communication, processing, and more. There are two types of computer hardware: external and internal.



Types of Computer Hardware



Internal Devices

Internal describes a device that's installed inside the computer. For example, a graphics card is an internal device.

External Devices

External describes a hardware device that's installed outside of the computer. For example, a printer.

Internal hardware examples
★ Motherboard ★ RAM ★ Drive (e.g., Blu-ray, CD-ROM, DVD, floppy drive, hard drive, and SSD). ★ Network card ★ Sound card ★ Modem

External hardware examples
★ Monitor ★ CPU ★ Mouse ★ Keyboard ★ Printer ★ Speakers

FUNCTIONS OF KEYBOARD AND MOUSE

The Keyboard and the Mouse are used to enter data into the computer; hence they are called input devices.

Keyboard



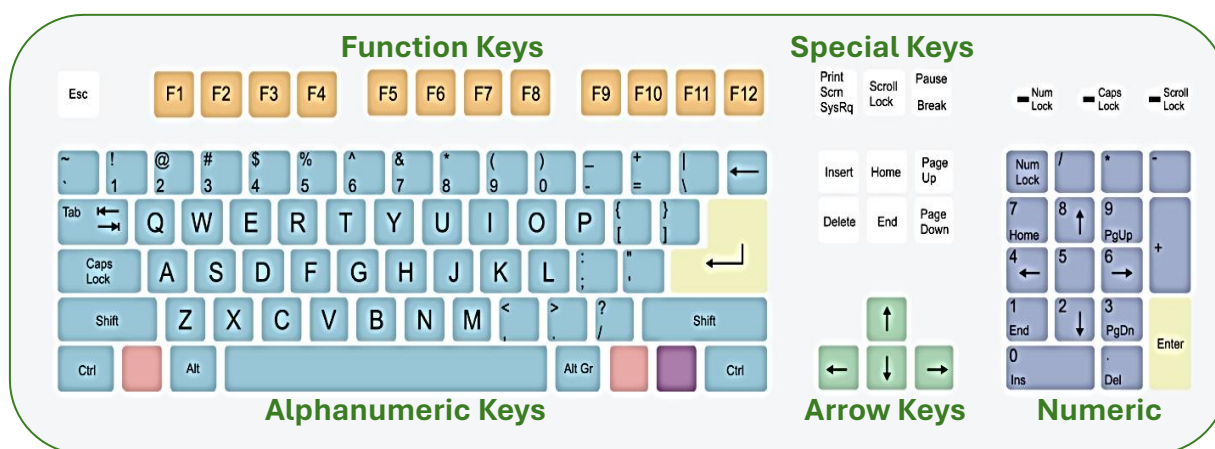
Keyboard is an input device which is similar to typewriter.

Keyboard has a set of keys which allows the user to give input to the computer in the form of letters, numbers and symbols. The standard keyboard has 104 keys approximately.



The keys in the keyboard can be grouped under five categories.

- ★ Letter keys (A to Z)
- ★ Number keys (0 to 9)
- ★ Arrow keys
- ★ Function keys (F1 to F12)
- ★ Special keys such as Enter, Esc, Delete etc.



DO YOU KNOW???

Spacebar key is the longest key on computer keyboard.



Mouse



The mouse is a device that allows the cursor to move around the screen. It is used to perform the following functions:



- ★ Point
- ★ Select
- ★ Click
- ★ Scroll
- ★ Drag-and-drop

Actions of a Mouse

- ★ **Left Click** - Pressing and releasing the left mouse button once.
- ★ **Double click** - Pressing and releasing the left mouse button twice in quick succession.
- ★ **Right click** - Pressing and releasing the right mouse button once.
- ★ **Drag** - Moving the mouse while keeping the left mouse button pressed.
- ★ **Drop** - Releasing the left mouse button after dragging it.



The First Computer Mouse



DO YOU KNOW???

Doug Engelbart invented the computer mouse in 1964. It was made of wooden shell.



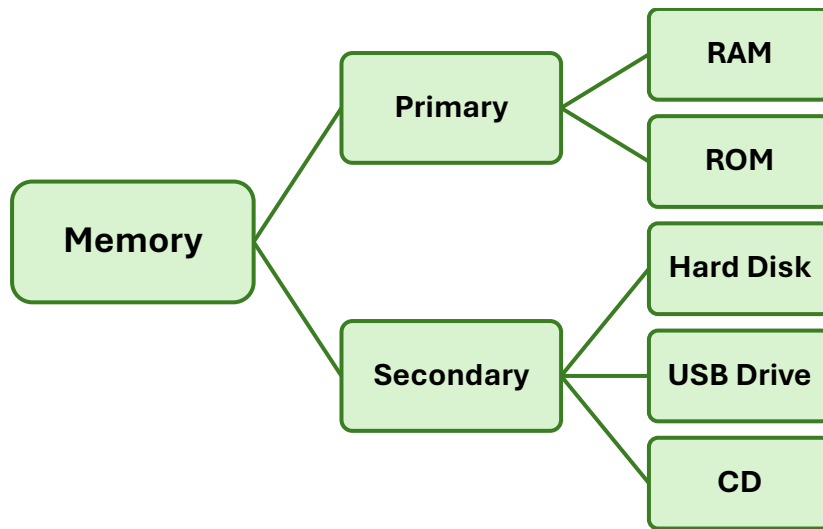
COMPUTER MEMORY & ITS TYPES

A memory is just like a human brain. It is used to store data and instructions. Computer memory is the storage space in the computer, where data to be processed and instructions required for processing are stored.



Memory is of two types:

- ★ Primary Memory
- ★ Secondary Memory

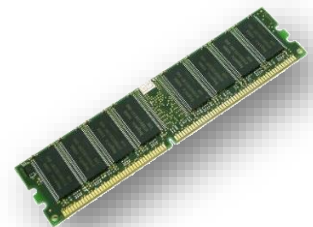


PRIMARY MEMORY (MAIN MEMORY)

Primary memory holds only those data and instructions on which the computer is currently working. It has limited capacity and comes in two types: volatile and non-volatile.

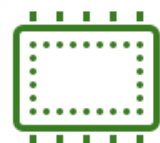
Types of Primary Memory

RAM (Random Access Memory) - RAM is a computer's short-term memory (volatile), where the data that the processor is currently using is stored.

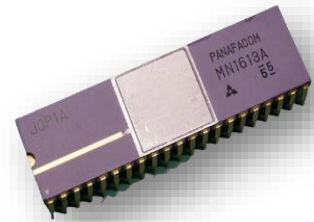


DO YOU KNOW???

Volatile memory requires power to retain data, whereas non-volatile memory retains data even without power.



ROM (Read Only Memory) - ROM (Read Only Memory) is a non-volatile memory type. This means it receives data and permanently writes it on a chip, and it lasts even after you turn off your computer.



SECONDARY MEMORY

Secondary memory can be internal or external and is a non-volatile memory. It is slower than the primary memory. These are used for storing data/information permanently. Examples: Optical discs, Floppy disks, Pen drives (USB drives), Hard disk, Memory card etc.

PRINTER AND ITS TYPES

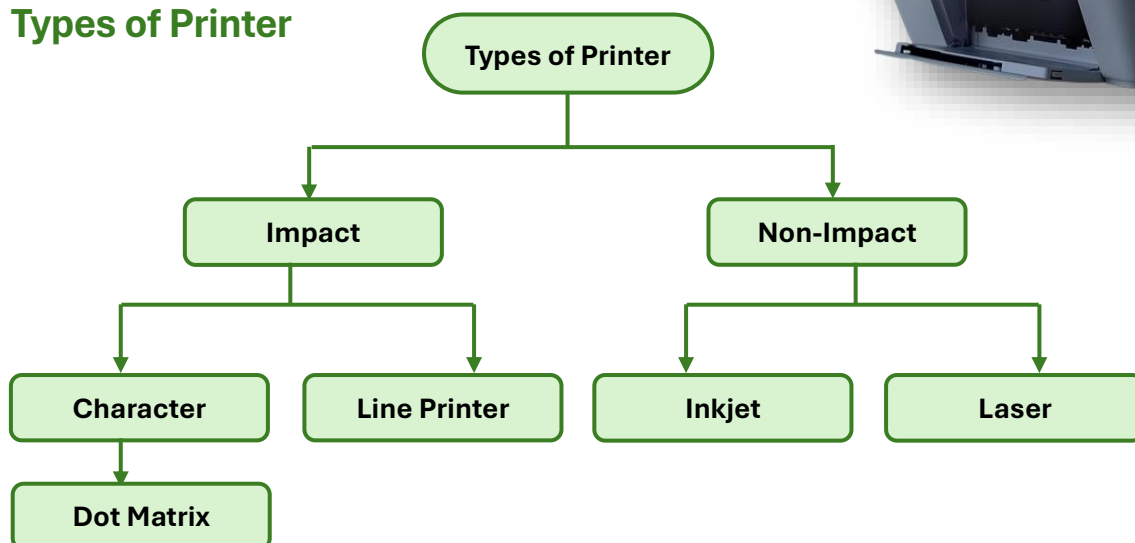
Printers are the most commonly used and popular output devices. They provide information in a permanent readable form. They produce printed output of results, programs and data.



Prints text or image on paper called hard copy. The printing speed of a printer is measured in pages per minute (ppm). Printer resolution is often measured in dots per inch (dpi).



Types of Printer



Printers can be divided into two main categories:

Impact Printers

It uses hammers or pins strike against a ribbon and paper to print the text. This mechanism is known as electro-mechanical mechanism. There are two types of printer : Character Printer & Line Printer

★ **Character Printer :** It prints only one character at a time. It has relatively slower speed. e.g. Daisy wheel printer. Dot matrix printer is also a character printer which prints characters as combination of dots. These have a matrix of pins on the print head of the printer which form the character.



★ **Line Printer:** The line printer is an impact computer printer that prints one entire line of text at a time.



Non-Impact Printers

Non-impact printers create figures and pictures without any connection between the printing device and the paper.

★ **Ink-Jet Printer:** It prints characters by spraying patterns of ink on the paper from a nozzle or jet.

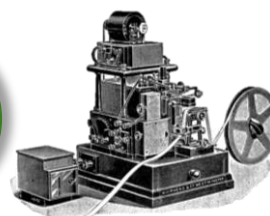


★ **Laser Printer:** It utilizes a laser beam to produce an image on a drum. The light of the laser alters the electrical charge on the drum wherever it hits.



DO YOU KNOW???

Syphon Recorder is the first Inkjet Printer (1867).



Let's Recapitulate



- ★ Hardware refers to the external and internal devices and equipment that enable you to perform major functions such as input, output, storage, communication, processing.
- ★ There are two types of computer hardware: external and internal.
- ★ Keyboard has a set of keys which allows the user to give input to the computer in the form of letters, numbers and symbols.
- ★ The mouse is a device that allows the cursor to move around the screen.
- ★ A memory is just like a human brain. It is used to store data and instructions.
 - ★ Primary Memory
 - ★ Secondary memory
- ★ Types of Primary Memory
 - ★ RAM (Random Access Memory)
 - ★ ROM (Read Only Memory)
- ★ Secondary memory can be internal or external and is a non-volatile memory.
- ★ Printers are the most commonly used and popular output devices. They provide information in a permanent readable form.

Let's Practice



I. Fill in the blanks with correct options

(Monitor, External, Main Memory, Power Supply, Internal, icons.)

1. Switch ON theto start the computer.
2. Hardware refers to the and..... devices.
3. is the example of external hardware device.
4. Primary memory is also known as
5. The small pictures on the desktop are called

II. Say true or false

1. We use drag option by mouse to select/shift the desired content.
2. Space key is the shortest key in keyboard.
3. ROM is a computer's short- term memory.
4. Hard Disk component of a computer is responsible for storing data permanently.
5. We use printer to get printed copies from the computer.

III. Multiple Choice Questions

1. ROM is _____
 a) Primary Memory b) Different Memory
 c) Secondary Memory d) None of these
2. is a Storage device
 a) CD b) DVD
 c) Hard disk d) All of these
3. Which of the following is not hardware?
 a) Monitor b) Mouse
 c) Operating system d) CPU
4. What do we use to move the cursor on the computer screen?
 a) Monitor b) Keyboard
 c) Mouse d) Printer
5. Which part of the computer looks like a television screen?
 a) Keyboard b) Monitor
 c) Mouse d) CPU

IV. Short answer type questions

1. Write the steps to power-on a computer?
2. What is Computer Hardware?
3. Tell me any four examples of External Devices.
4. What are the types of printer?
5. What is ROM?

Let's Do Hands on



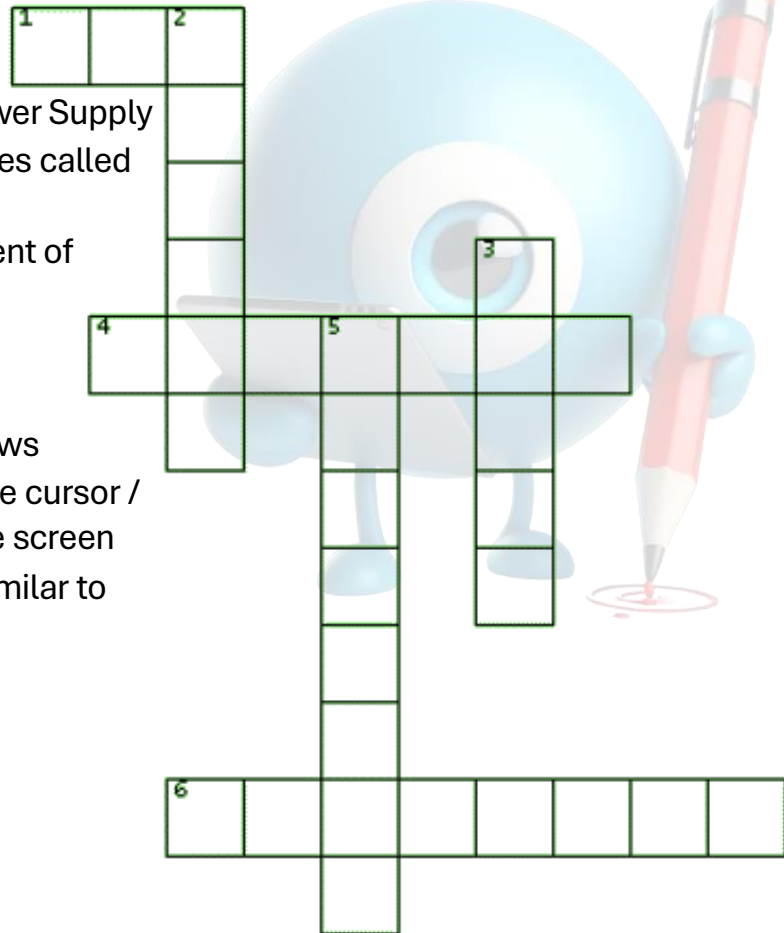
I. Solve the crossword puzzle using the hint(s) given below:

➡ Across

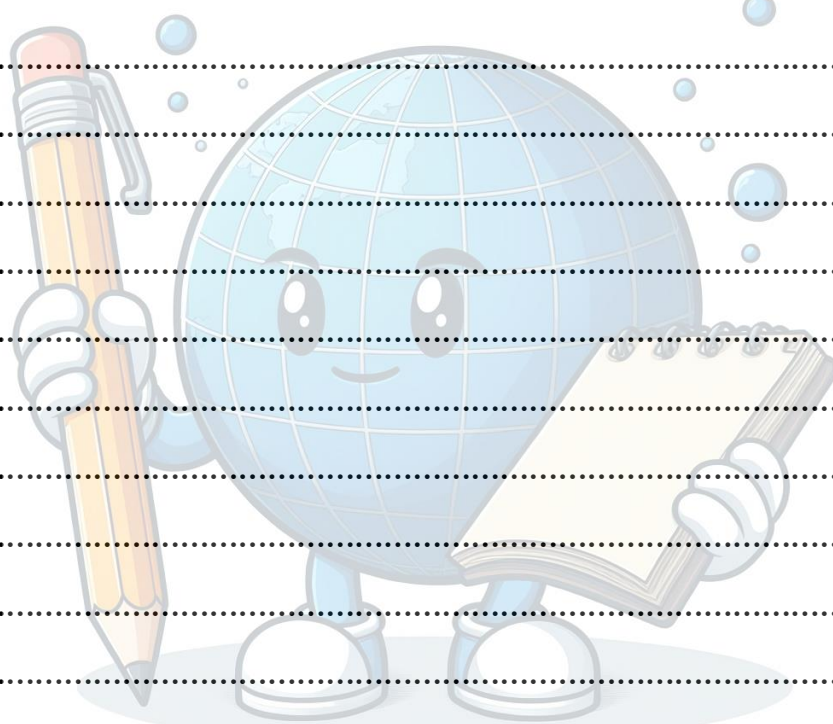
1. Uninterrupted Power Supply
4. It has small pictures called icons
6. Physical component of computer

⬇ Down

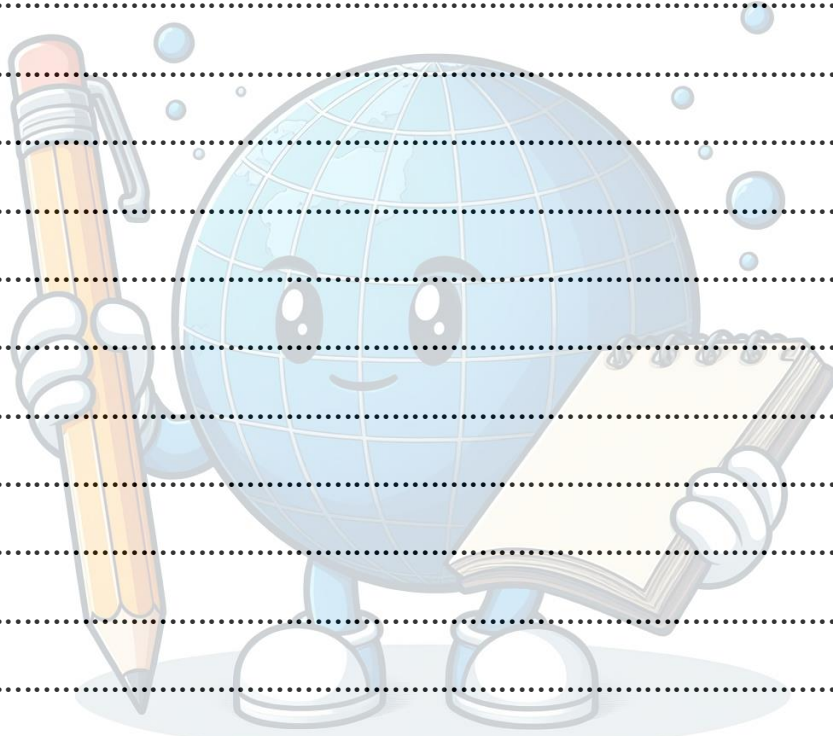
2. Desktop in Windows
3. Device to move the cursor / pointer around the screen
5. An input device similar to typewriter



Space for Notes



Space for Notes



Answer Key



Let's Practice

I. Fill in the blanks

1. Power Supply 2. External, Internal 3. Monitor 4. Main Memory 5. Icons

II. Say true or false

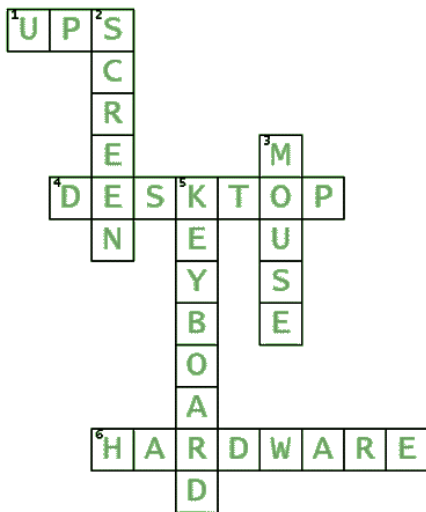
1. True 2. False 3. False 4. True 5. True

III. Multiple choice questions

1. a) 2. d) 3. c) 4. c) 5. b)

Let's Do Hands on

I. Crossword puzzle



INTRODUCTION TO WINDOWS

Learning objectives

In this chapter, you will be able to understand:

- ★ Operating System
- ★ Windows and Changing Screen Saver
- ★ Parts of a Window
- ★ Minimize, Maximize, Close, Resize and moving a window
- ★ Understand the concept of files and folders and using accessories
- ★ Starting an application using Program Menu and the run option
- ★ Performing calculation using Calculator



INTRODUCTION

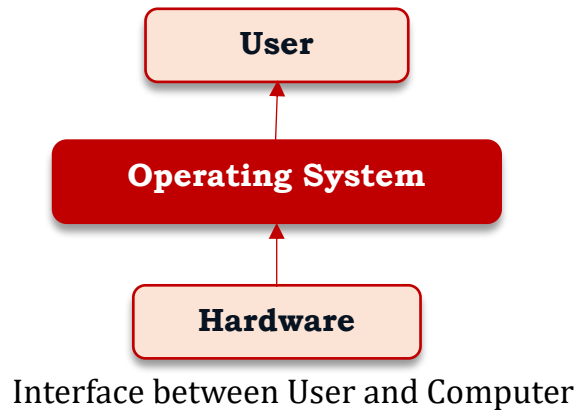
In the previous chapters we have studied about basics of computer which included parts of computers, its characteristics, input and output devices, storage & memory, desktop, etc. where the students were able to understand the fundamental concepts and components of Computers.

In this chapter students will be able to learn about the features of an Operating System.



WHAT IS AN OPERATING SYSTEM

An Operating System (OS) is an interface between a user and the hardware.



The three most common operating systems for personal computers are Windows, macOS, and Linux.

You may be familiar with Windows. Let us learn about it.

WINDOWS

Windows is an operating system designed by Microsoft. Windows comes preloaded on most new personal computers (PCs), which helps to make it the most popular operating system in the world.



Different versions of Windows

- ★ Windows Vista (2006)
- ★ Windows 7 (2009)
- ★ Windows 8/8.1 (2012-2013)
- ★ Windows 10 (2015)
- ★ Windows 10 S (2017)
- ★ Windows 11 (2021)

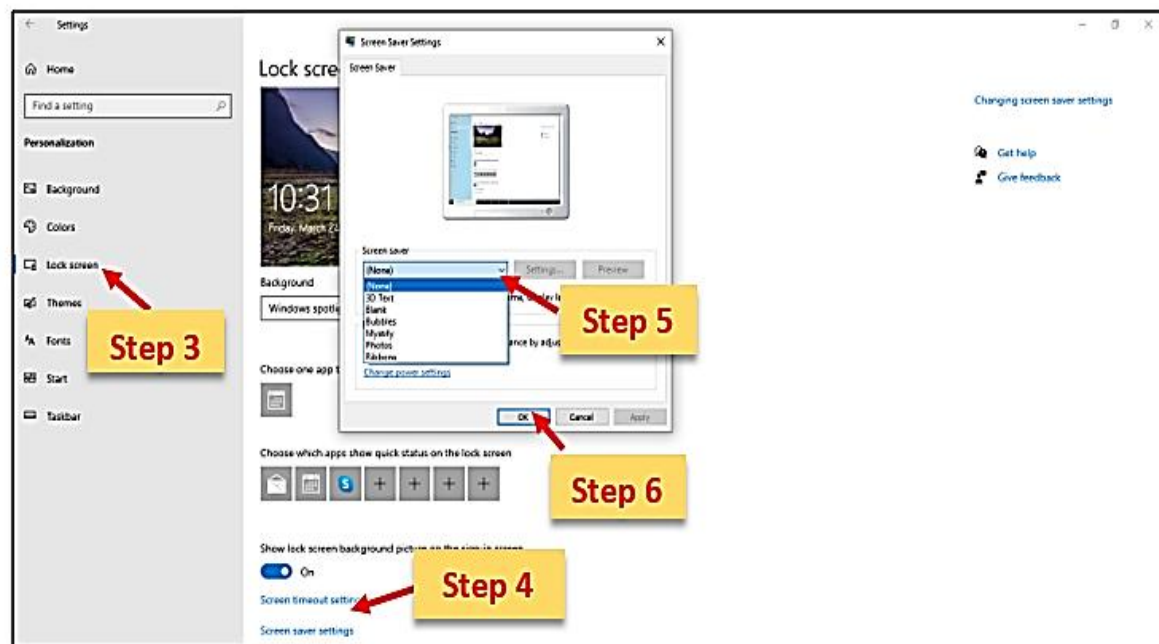
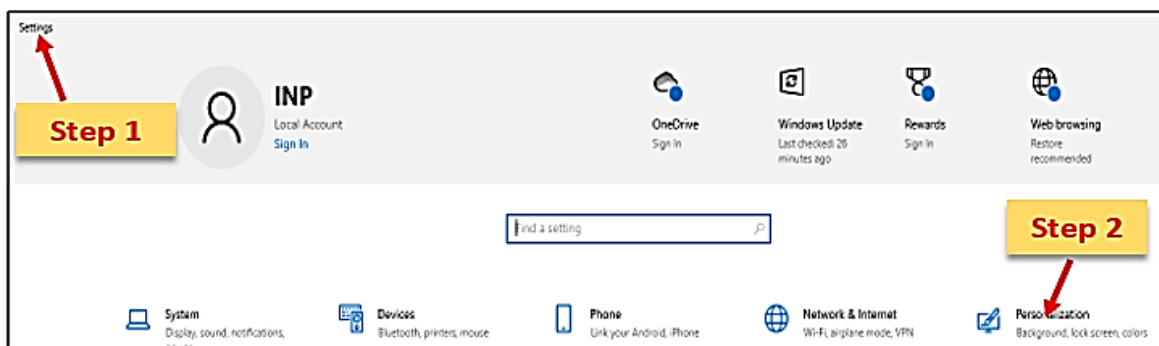
SCREEN SAVER

A screen saver is a software program that becomes activated after the computer is inactive for a specified time. It displays moving pictures or graphics on the computer. Screen saver prevents viewing of desktop contents while the user is away.

Changing Screen Saver

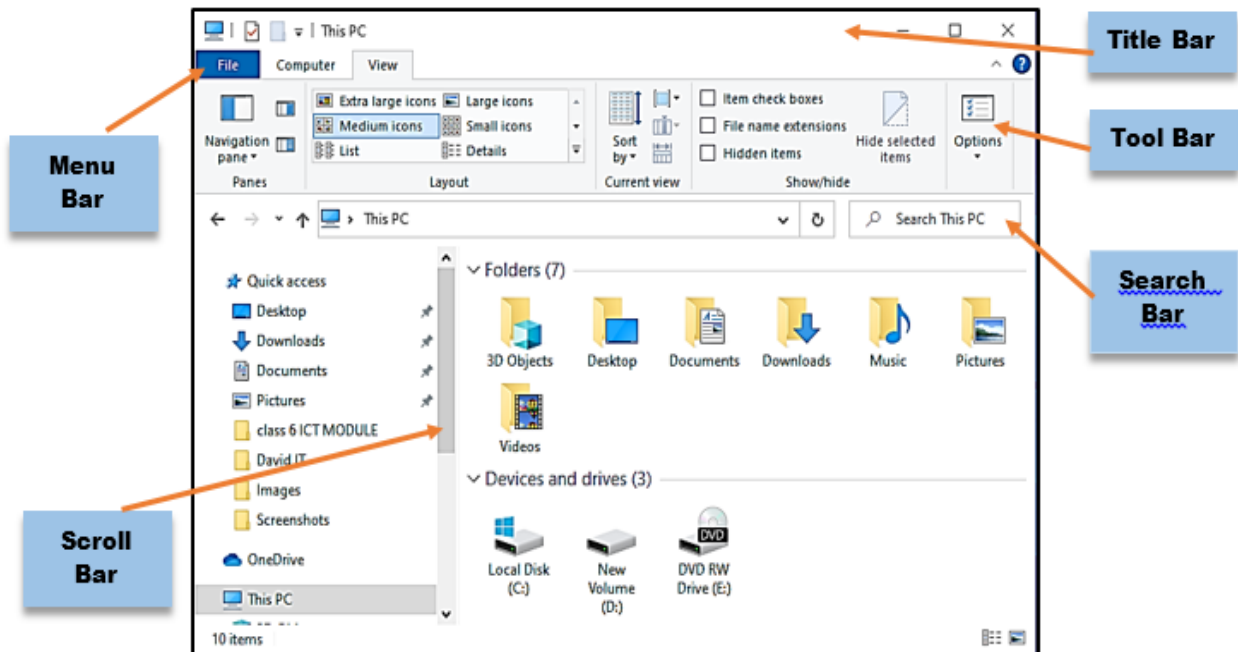
You can easily change your PC's Screen Saver by following the given procedure:

Go to Settings > Personalization > Lock screen > Screen saver settings > Select Screen saver > Ok.



PARTS OF A WINDOWS

When you click an icon on the desktop, a window is opened. For example, when you click 'This PC' icon, a window opens. The parts of the window are - Title bar, Menu bar, Tool bar, Search Bar and Scroll Bar.



Title bar :

The horizontal bar at the top of a window that contains the name of the window is called Title bar.

Menu bar :

The horizontal bar found just below the title bar, which contains many options, is called the menu bar. When these options are clicked, a list of commands appears with which you can perform various functions.

Tool Bar :

The toolbar gives shortcuts for commonly used functions listed in the menu bar.

Search Bar :

Search Bar is used to search a particular item.

Scroll Bars:

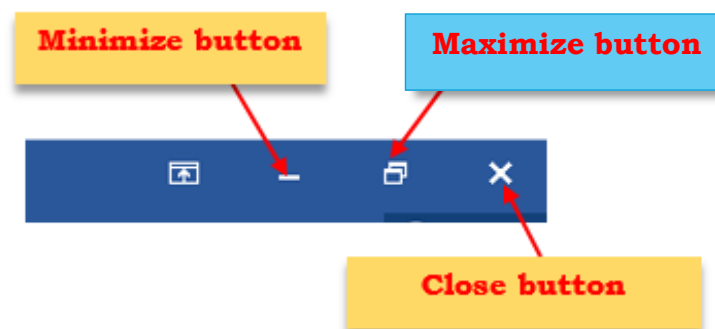
Scroll bars helps to scroll through the contents of a window. The scroll bar appears if the window is too small to show the entire contents of the work area.

A window can consist of two scroll bars:

- ★ Vertical scroll bar – It is used to move through a window vertically
- ★ Horizontal scroll bar- It is used to move through a window horizontally.

MINIMIZE, MAXIMIZE AND CLOSE A WINDOW

The three buttons on the right side of the title bar are - **Minimize button**, **Maximize button** and **Close button**.



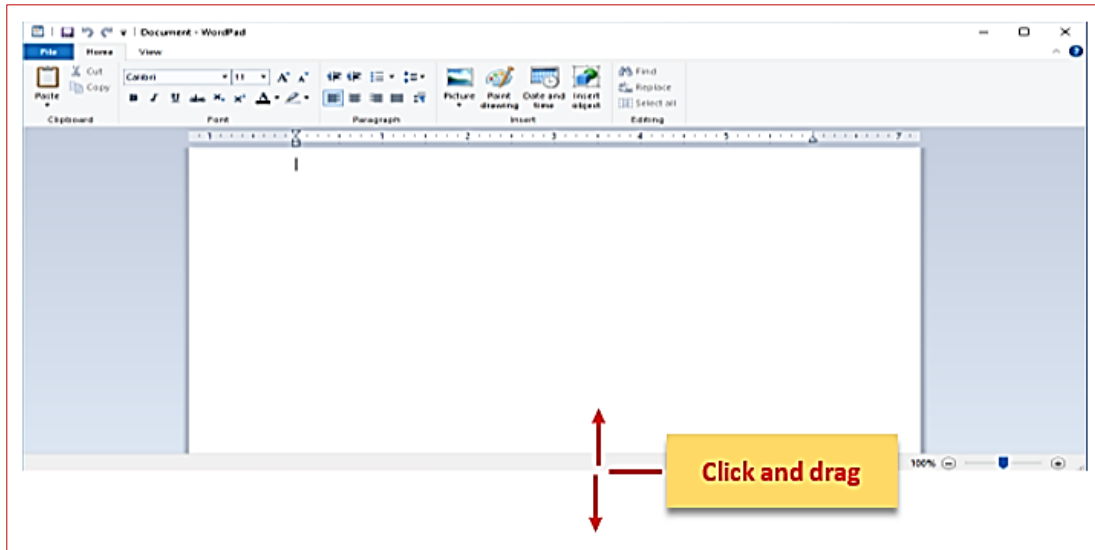
1. The Minimize button minimizes the window to a button on the taskbar. To restore the window, click its button on the taskbar.
2. When you click on the Maximize button, the window occupies the whole screen. On maximizing a window, the Maximize button changes to the Restore button.
3. To bring it back to its original size, click the Restore button. On restoring the window, the Restore button will be replaced by the Maximize button.
4. To close a window, click the Close button.

RESIZE AND MOVE A WINDOW

Resize a Window : It means to change the height, size or width of a window.

How to resize a window:

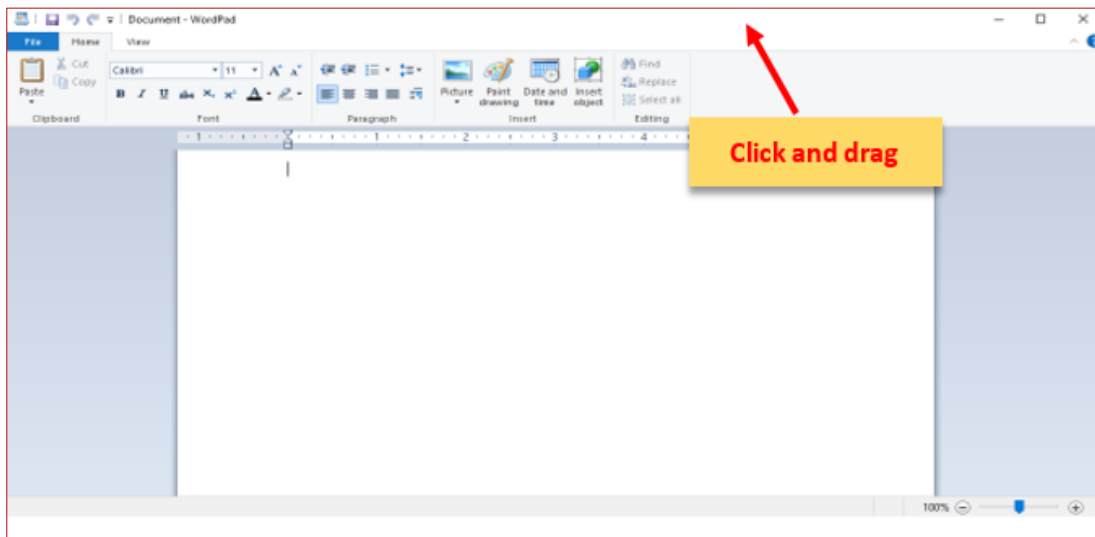
Click on the restore button (when the window is maximized) and drag the window at any of the window borders until it is at the desired size.



Moving a Window : It means changing the position of a window in the computer screen.

How to move a window:

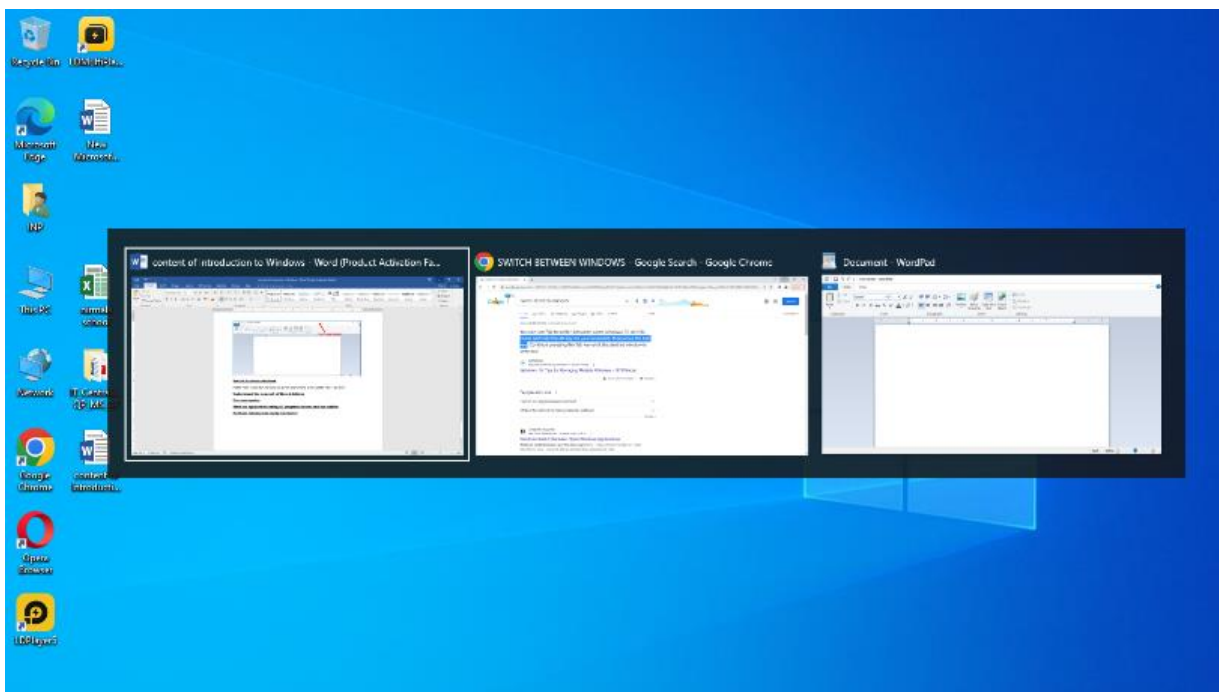
Click and drag it at the title bar to a new location.



SWITCH BETWEEN WINDOWS

Sometimes we work on multiple windows by keeping all of them open in PC and all the windows are displayed as buttons on the taskbar. The button of the windows which are currently being used are highlighted on the taskbar and are called active windows.

To switch to some other windows, click its button on the taskbar. This window will appear on the desktop and its button will get highlighted on the taskbar. The other way of switching to a window is by pressing Alt + Tab.



FILES & FOLDERS

Folder

A folder is a small yellow icon. A folder is used to store files on a particular topic together in a common location.

You may think that files can be stored directly in the hard disk. So what is the use of folders?



Imagine that you have to search for a book in a library that has no racks. Books of different subjects are put together. You will find it very difficult to get the book you want.

Similarly, if all the files in a computer are stored together it becomes difficult to search for a particular file. So, it is important to organize the different files in your computer. For this, Windows provides you with folders.

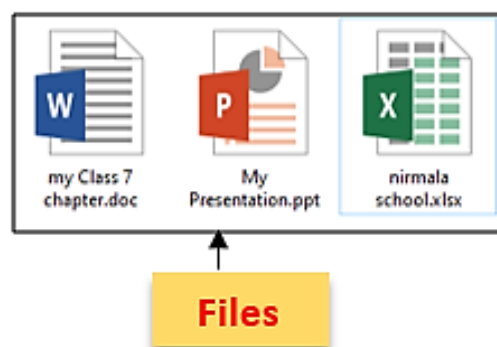
A folder can contain files. It can also contain other folders in it. These are called subfolders.

Files

A file is a resource in computer to store information.

Every one of you has a name. Similarly, each file must have a name to identify it. This is called the file name.

All files have two parts in their names- the Primary name and the Extension.

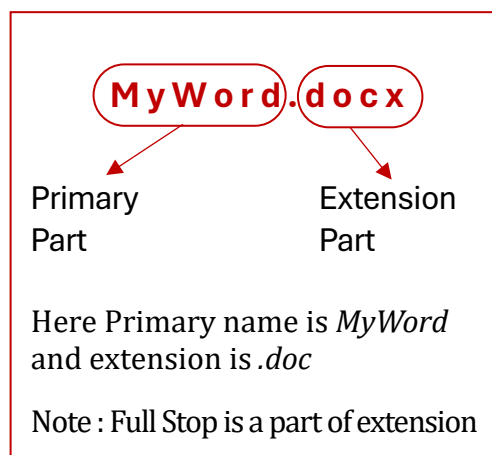


★ Primary name – This is the first part of the filename. It is the name by which a file is identified.

★ Extension – The second part contains the extension of the file name. This identifies the type or family of the file.

For example, picture files have the extension .bmp and text files have the extension .txt. An example for a file name is MyWord.doc

A file name can contain- Alphabets, Numbers, Spaces and Special characters, such as [] { }, ! \$ ~ ' % @



DO YOU KNOW???

Bill Gates and **Paul Allen** founded Microsoft and Windows operating system.





ACCESSORIES

Paint | Notepad | Wordpad | Media Player | Calculator | Games

Paint

It is used to draw, colour and edit pictures like crop, remove, resize, rotate etc. **Click Start Button > Paint.**

Notepad

It is a simple text editing program which enables computer users to create a document. **Click Start Button > Notepad.**

Wordpad

It can format and print text in a variety of styles, including bold, italic, coloured, and centered text, but it lacks features like a spell checking, thesaurus, and pagination control. **Click Start Button > Wordpad.**

Media

It is a software that plays audio, video or animation files in the computer. **Click Start Button > Media Player.**

Calculator

It is a software that performs arithmetic operations like addition, subtraction, multiplication, division etc. **Click Start Button > Calculator.**

Games

It is a software program that enable a player to interact with a virtual game environment for entertainment and fun using computer input devices like keyboard, monitor etc. **Click Start Button > Game.**

START AN APPLICATION

Using ALL PROGRAMS menu and RUN Option

There are two ways to start an application in Windows:

- ★ Using the All Programs menu
- ★ Using the Run option on the Start menu

Using the All Programs menu

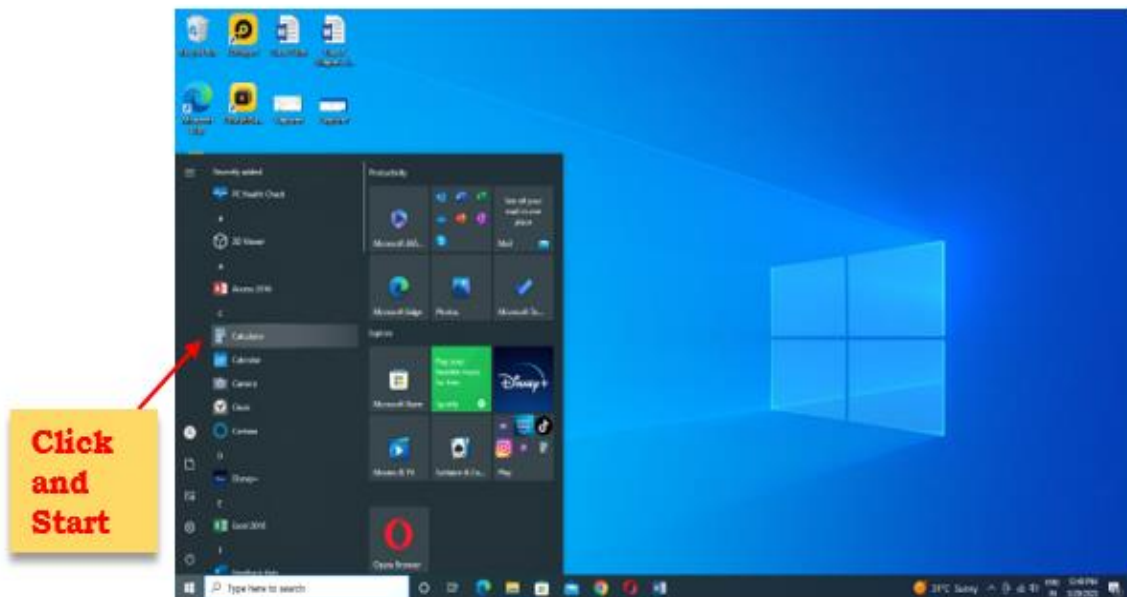
The applications in Windows can be started using the **All Programs** on the **Start menu**.

For example, there is an application called Calculator that is used for making calculations.

To open Calculator through the All Programs menu:

1. Go to Start Button
2. Click on the name of application i.e. 'Calculator' from the list.

The Calculator opens. You can now perform calculations using it.



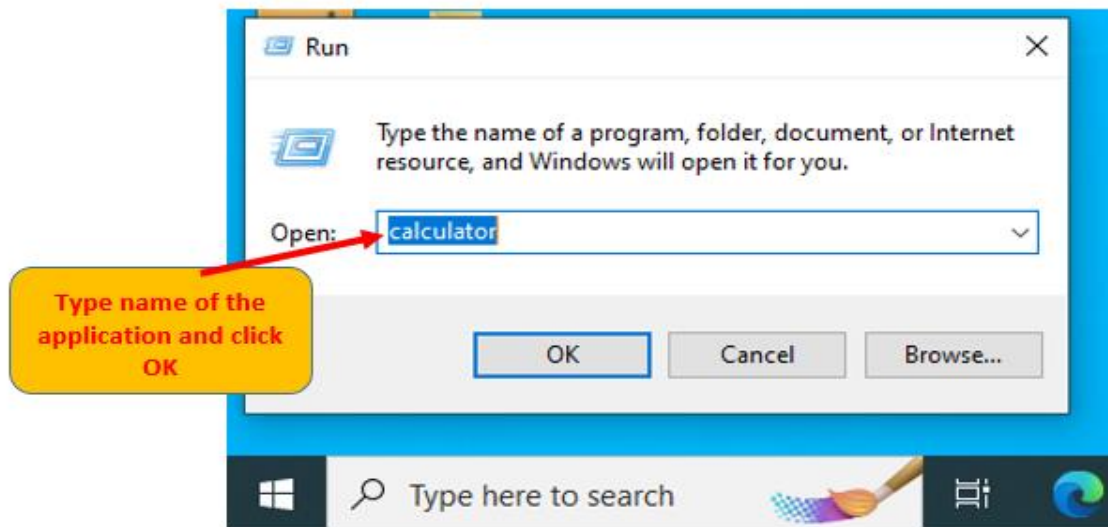
Using the Run Option

The Run option on the Start menu can also be used to open or start a program.

To open the Calculator program through the Run option:

1. Right click the Start button.
2. Click the Run option on the Start menu. The Run dialog box appears.
3. In this dialog box, type the name of the program that has to be opened in the Open box.

For example, type calc to open calculator.

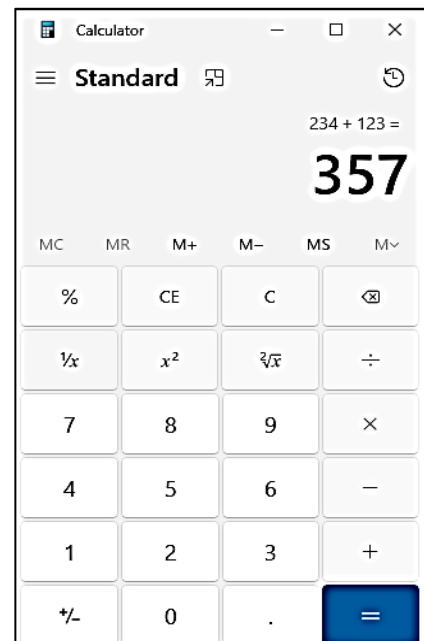


Perform calculations using Calculator

Like our pocket calculator, windows is also having a calculator to perform calculations, such as addition, subtraction, multiplication, division, square root etc. You can use keyboard of the Windows Calculator, to type the numbers or the mouse to click the number buttons.

For example, to add 234 to 123:

1. Click the number buttons 2, 3 and 4.
2. Click the button +.
3. Click the number buttons 1, 2 and 3.
4. Click the button =



Now the sum of the two numbers, 357, is shown in the result text box.

Let's Recapitulate



- ★ An Operating System (OS) is an interface between the user and the hardware.
- ★ Windows is an operating system designed by Microsoft. Windows comes preloaded on most new personal computers (PCs), which helps to make it the most popular operating system in the world.
- ★ A screen saver is a software program that becomes activated after the computer is inactive for a specified time.
- ★ A folder is a small yellow icon. A folder is used to store files on a particular topic together in a common location
- ★ A file is a resource in computer to store information. Each file must have a name to identify it. This is called the file name.

Let's Practice



I. Fill in the blanks

1. An _____ is an interface between a computer user and the computer hardware.
2. Windows is an operating system developed by _____.
3. _____ displays moving pictures or graphics on the computer when system is idle for a specified time.
4. A folder is a small _____ colour icon.
5. Information is stored in a _____.

DO YOU KNOW???

First screensaver was written for the original IBM PC by John Socha in **1983**.



II. Say True or False

1. Windows is also having a calculator to perform calculations.
2. .bmp is an extension of word document.
3. Screen Saver is a hardware installed in the mother board.
4. Media player is used for store the data.
5. The Run option is used to open or start a program.

III. Multiple Choice Questions

1. Windows developed by _____
a) Apple b) Microsoft
c) IBM d) Google
2. Windows have inbuilt application.
a) MS Office b) VLC Media Player
c) Chrome d) Calculator
3. _____ is used for edit pictures.
a) Notepad b) Paint
c) Calendar d) Calculator
4. Media player is used for _____.
a) Video and Audio b) Drawing
c) Typing d) Printing
5. .docx/.doc is an extension for
a) Excel file b) image
c) Video d) word document

IV. Short answer type questions

1. What is an Operating System?
2. What are the steps to open Calculator application by using Run Option?
3. Name the parts of a window?
4. List the programs available in accessories option?
5. What are difference between file and folder?

Let's Do Hands on



I. Label the parts of the following figure.



II. Switch on your computer system and do the following.

1. Change desktop background.
2. Set a new screen saver
3. Set system date and time.
4. Create a new folder on the desktop and rename it with your name.

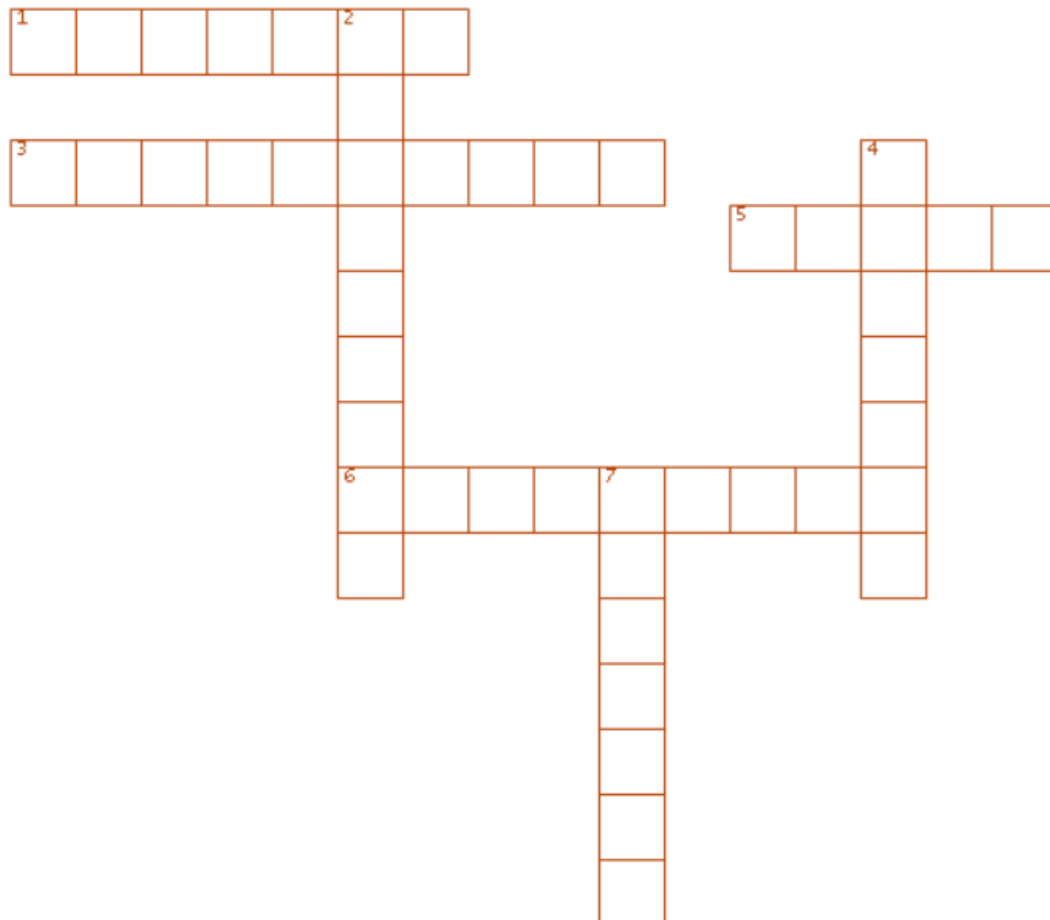
III. Solve the crossword puzzle using the hint(s) given below:

➡ Across

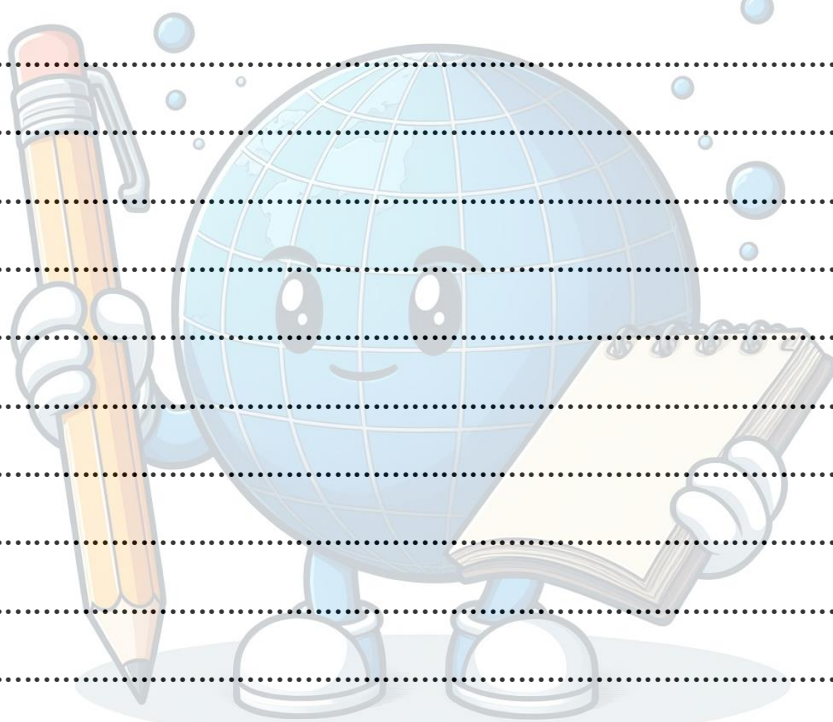
1. An OS developed by Microsoft Corporation
3. Windows program to perform calculations
5. Small pictures on Desktop
6. Part of filename after dot(.)

⬇ Down

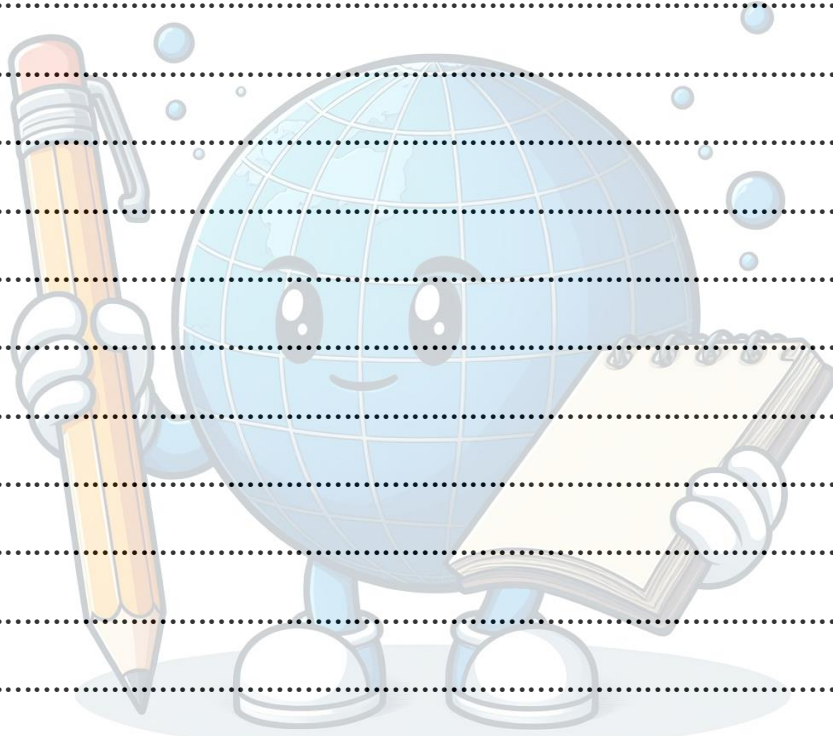
2. Image in the background
4. Loading OS into memory
7. Simple text editor



Space for Notes



Space for Notes



Answer Key



Let's Practice

I. Fill in the blanks

1. Operating System 2. Microsoft 3. Screen saver 4. Yellow 5. Hard Disk

II. Say true or false

1. True 2. False 3. False 4. False 5. True

III. Multiple choice questions

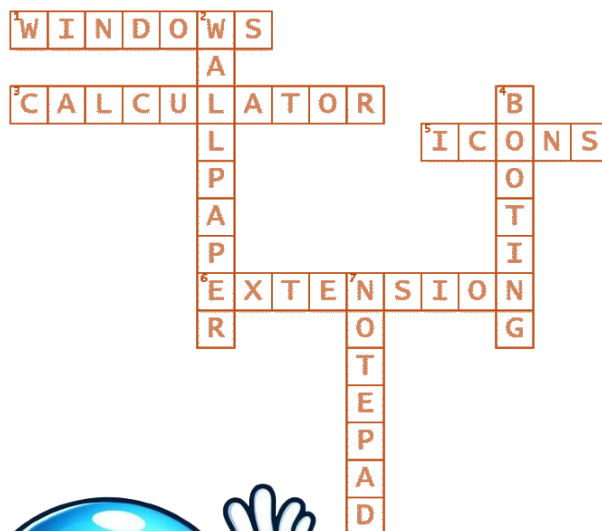
1. b 2. d 3. b 4. a 5. d

Let's Do Hands on

I. Label the parts of the following figure.

1. Start menu 2. Start button 3. Taskbar 4. Notification area

III. Crossword puzzle



MS-PAINT & PAINT 3D

Learning objectives

In this chapter, you will be able to understand:

- ★ Overview of MS-Paint.
- ★ Getting started with MS-Paint.
- ★ Getting familiar with tools-draw pictures, fill colors, use different shapes etc.
- ★ Creating and editing pictures
- ★ Understanding MS 3D Paint.
- ★ Work with basic tools and 3D objects in Paint 3D.
- ★ Applying Sticker, Magic Select and Grouping effects.



OVERVIEW OF MS- Paint

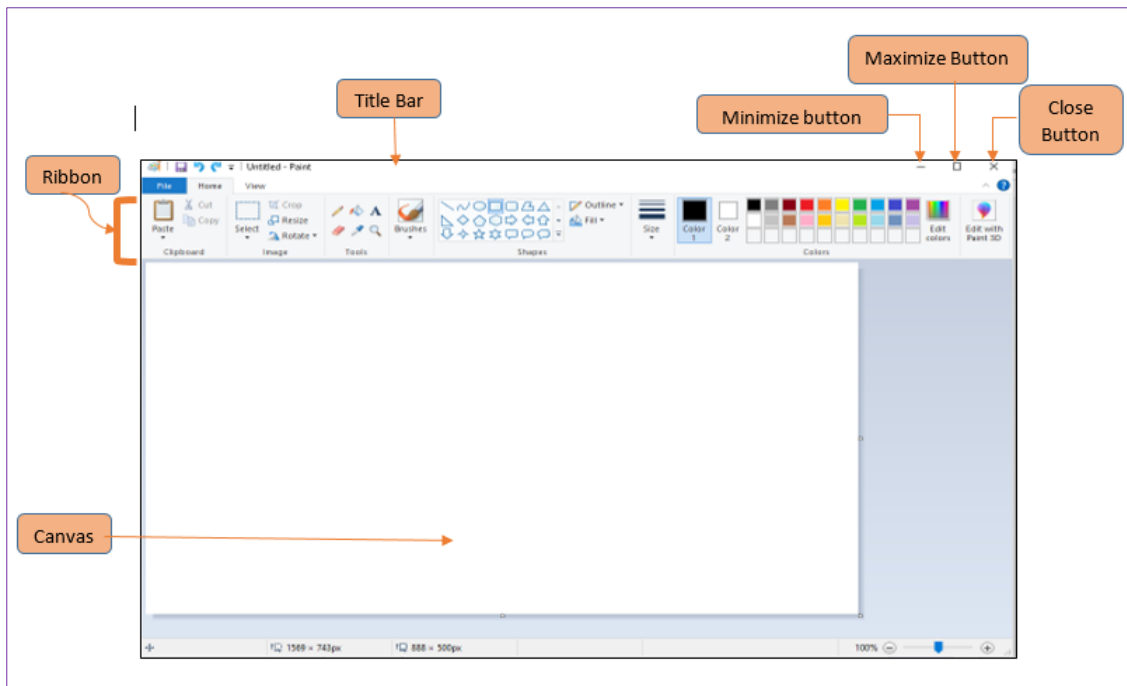
MS-Paint or Microsoft Paint is an application program included with Microsoft Windows. It can be used for drawing and painting purposes. It can also be used to edit and color existing pictures.

Getting started with MS-Paint

To open MS Paint, follow these steps:

Click Start → Windows Accessories → Paint





BASIC ELEMENTS OF MS-PAINT WINDOW

As you open a MS-Paint window, the following basic elements are available in it:

Title Bar: The title bar is located at the top of the MS Paint window. It shows the name/title of the current image or picture or the name by which the file is saved.

Minimize: This button is clicked to hide the Paint window temporarily from the screen.

Maximize: This button enlarges the Paint window if it is in small size.

Close Button: This button is clicked to exit Paint application.

Ribbon: A ribbon holds a set of commands which are arranged in groups. The Home ribbon opens up by default in the Paint program.

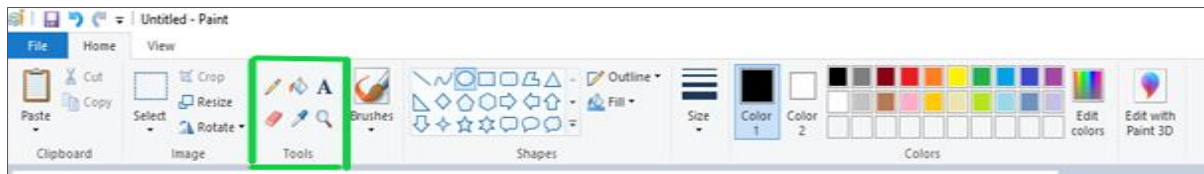
Canvas: A large white area where you draw and color is called canvas.

DO YOU KNOW???

First version of MS Paint' (called as Windows Paint') was programmed by **Dan McCabe**.



MS-PAINT TOOLS



Tools	Name	Description
	Pencil	The Pencil tool is used to draw free hand lines with a hard edge.
	Eraser	This tool is used to erase/remove the drawings, texts or shapes.
	Fill with color	This tool help to fill an area or object with selected color.
	Color Picker	By using this tool, Color is selected from one part of the object and can be used to color other areas. Left click is used for back color and right click for foreground color.
	Text	This tool is used to insert text in Paint.
	Magnifier	Magnifier is used to zoom in on parts of the picture.

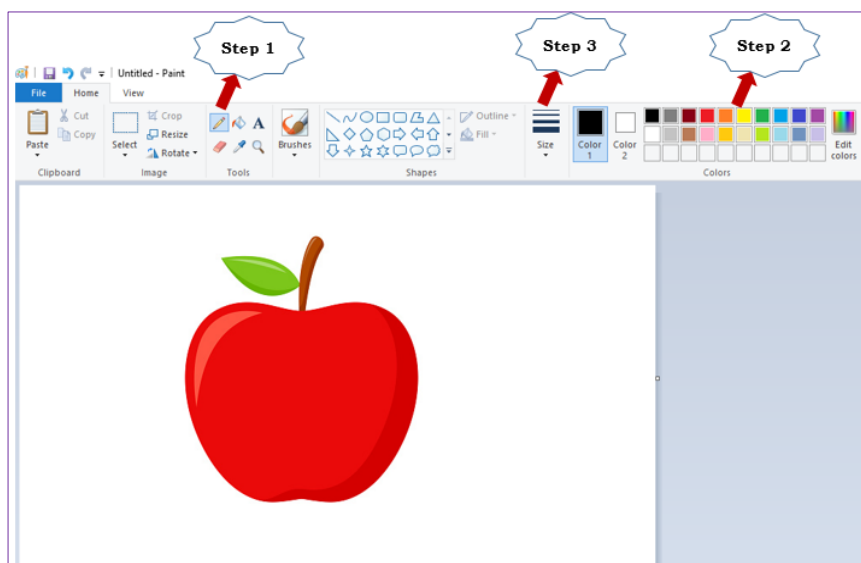
CREATE NEW PICTURES

To draw a picture, follow these steps.

Step 1: Click the Pencil Tool in the tool box.

Step 2: Select a color from the color box.

Step 3: You can change the thickness of the line by selecting the lines given in the Line Patterns.



DRAWING DIFFERENT SHAPES

The shape group in MS Paint hold many shapes like Line, Rectangle, Oval, Arrows, Callouts and many more. Let us learn to draw shapes in canvas.



Keyboard Shortcut

To cut, copy and paste an image :

Ctrl + X,
Ctrl + C
and
Ctrl + V

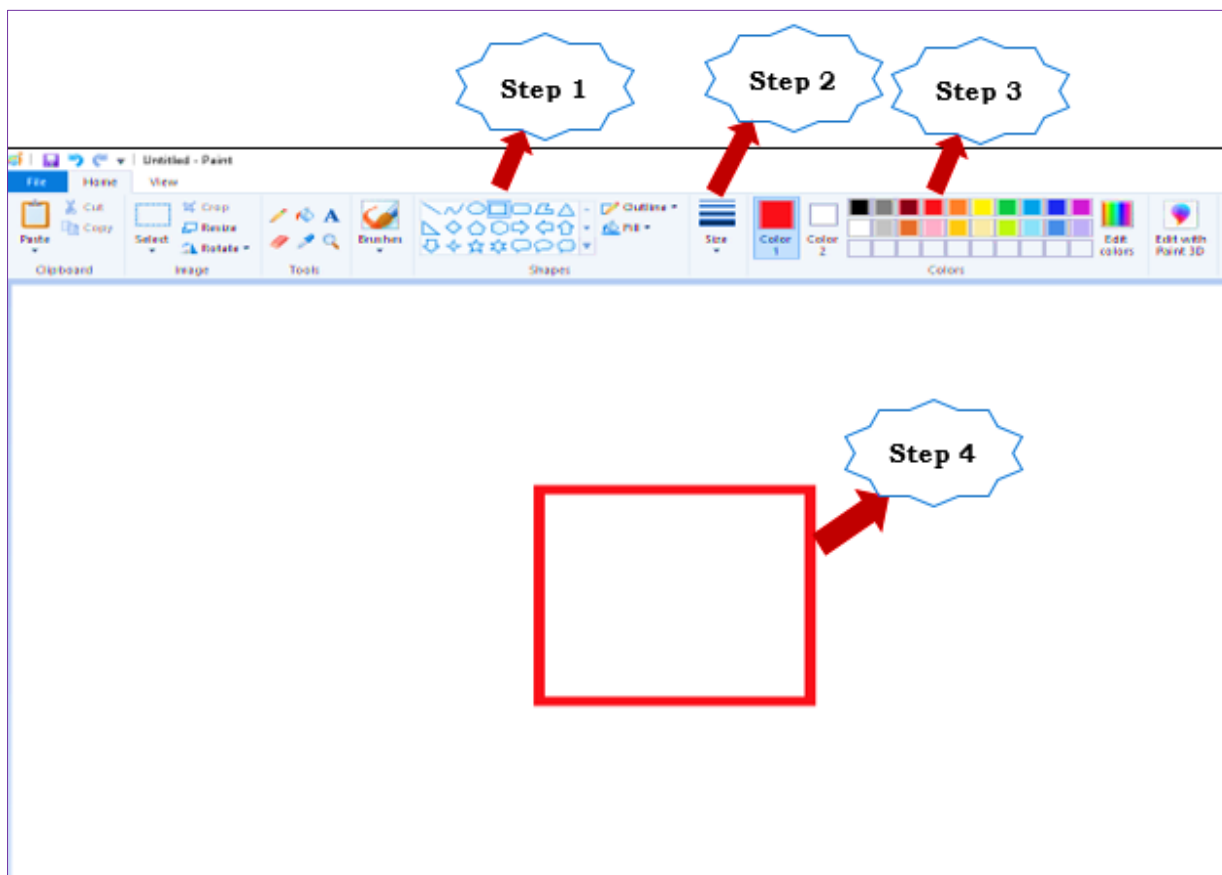
Steps to create different shapes:

Step 1: Click the Rectangle shape in the Shapes group.

Step 2: Click the Size option and choose thickness of shape outline.

Step 3: Choose a color from Colors group.

Step 4: On the canvas, click and drag the mouse to draw.



FILL COLOUR IN DRAWING

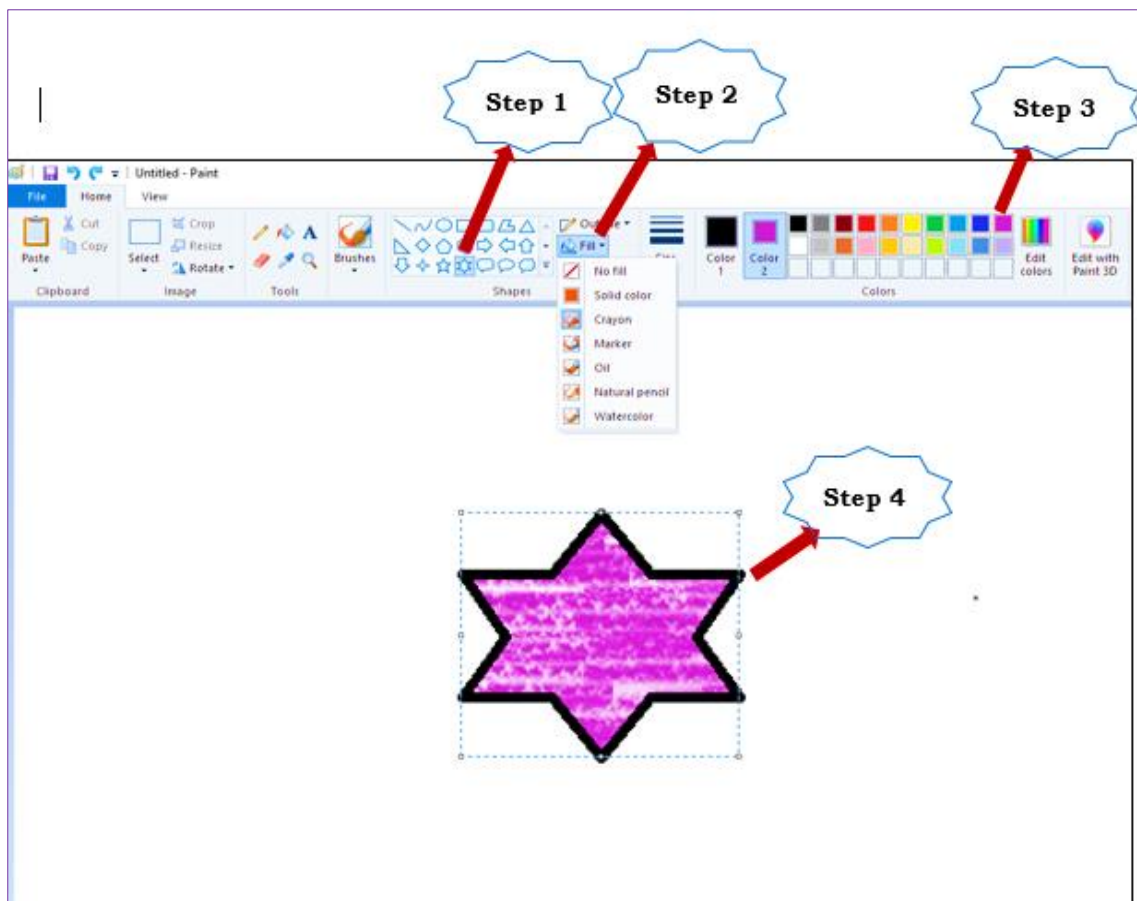
Fill with color tool is used to fill color inside the object. Follow these steps to fill your drawing with color:

Step 1: Select a shape to draw on the canvas.

Step 2: Click the Fill option and choose the required fill style.

Step 3: Click the Color 2 in the Colors group and select a color which is to be filled in a shape.

Step 4: On the canvas, click and drag the mouse to draw.



DO YOU KNOW???

Due to public demand, Paint was included with Windows 10 & 11.



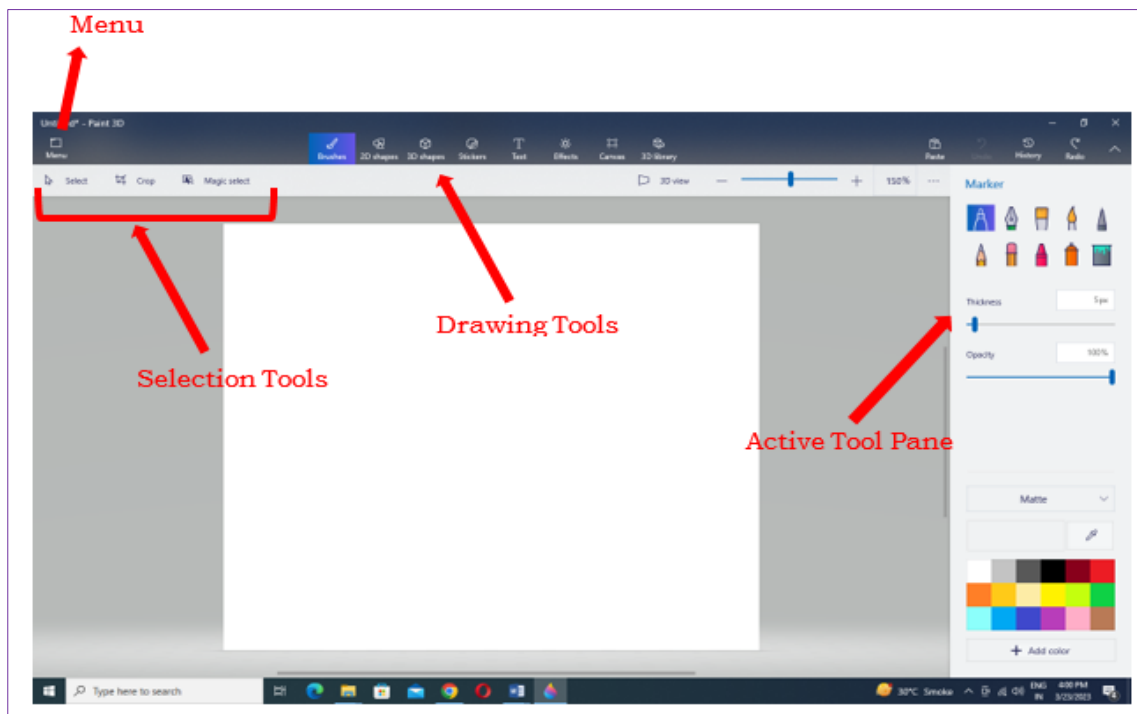
MS PAINT 3D – AN INTRODUCTION

Paint 3D is a free application program from Microsoft to create basic or creative objects by easily combining 2D and 3D tools.



Steps to Open Paint-3D

- ★ Click on Start
- ★ Select Paint 3D
- ★ Click on New to open a new project



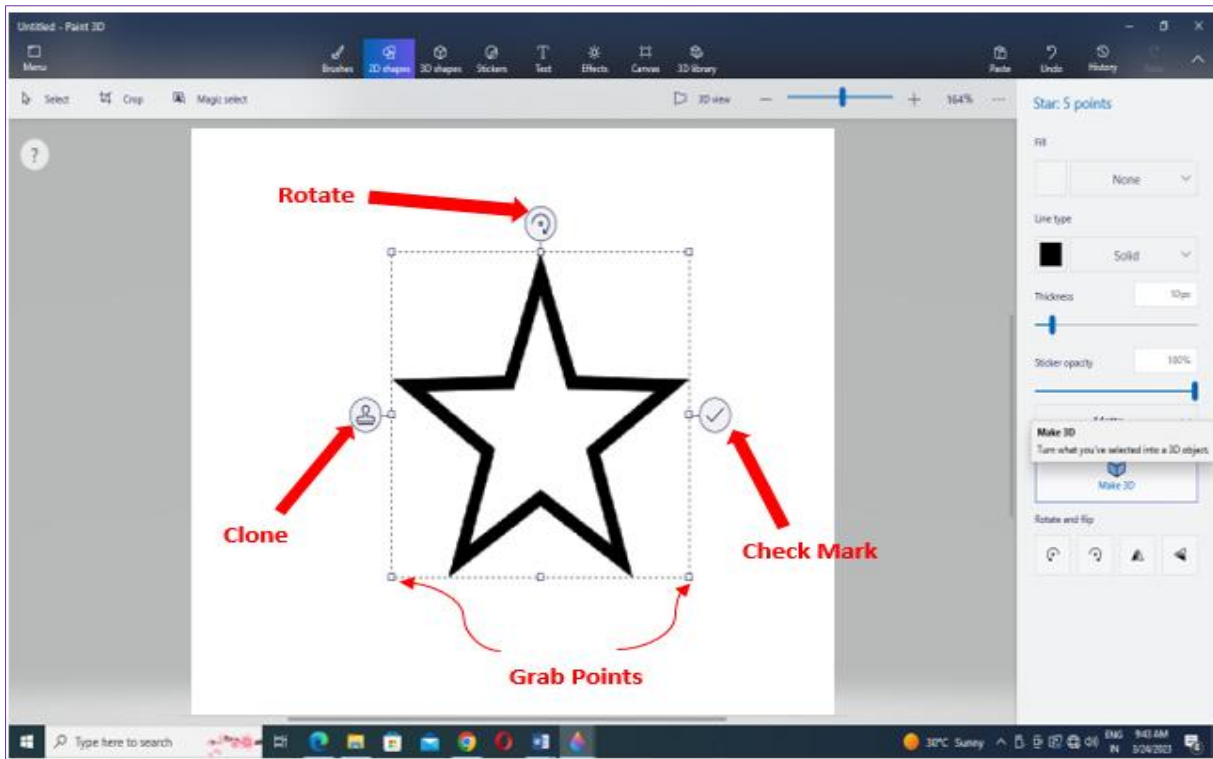
CREATING 2D SHAPES

Step1: Click on **2D Shapes**.

Step 2: From the Active Tool Pane select 2D Shape of your choice.

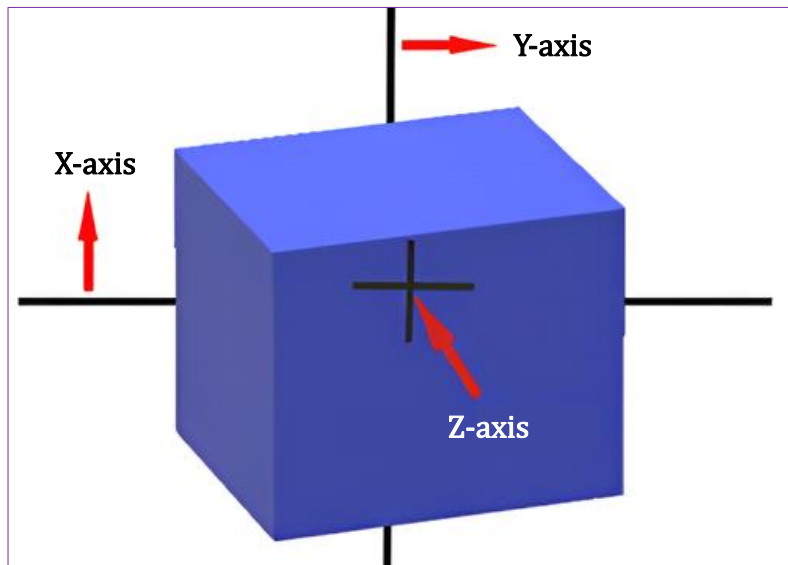
Step 3: Drag the shape on your drawing.

Grab Points at the corners can be used to change the size and proportions of the shape.
Rotate Tool can be used to rotate the shape.
Check Mark Tool completes the shape. You can also complete a shape by left clicking somewhere away from the shape.
Clone Tool can be used to make copy of the shape.



UNDERSTANDING PAINT 3D WORLD

The 3D world is typically visualized using three axes, X, Y, and Z. In Paint 3D, the X axis is horizontal, the Y axis is vertical, and the Z axis is in and out of the display. The Z axes appears as a point in the normal view. When you create a 3D object, there are tools for rotating the object around the three axes. When you draw 3D objects on the canvas, they can be moved in and out of the plane of the Canvas.



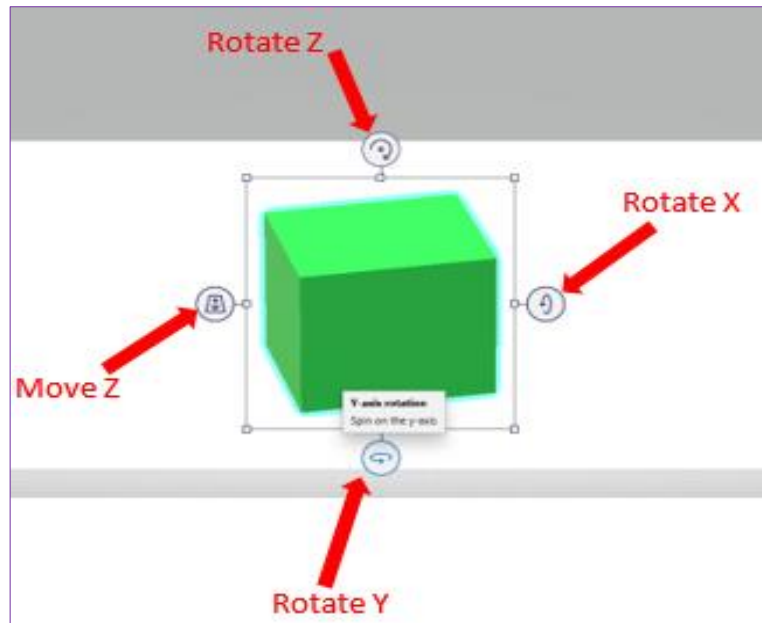
CREATING 3D SHAPES

3D Paint allows to create creative pictures with 3D Shapes. To add a 3D object, follow these steps:

- ✧ Select the 3D shapes tool
- ✧ Select the Cube object
- ✧ Drag out a cube

When a 3D object is created, tools are available to rotate the object around each of the 3 axes, and a fourth tool to move the object with respect to the plane of the Canvas.

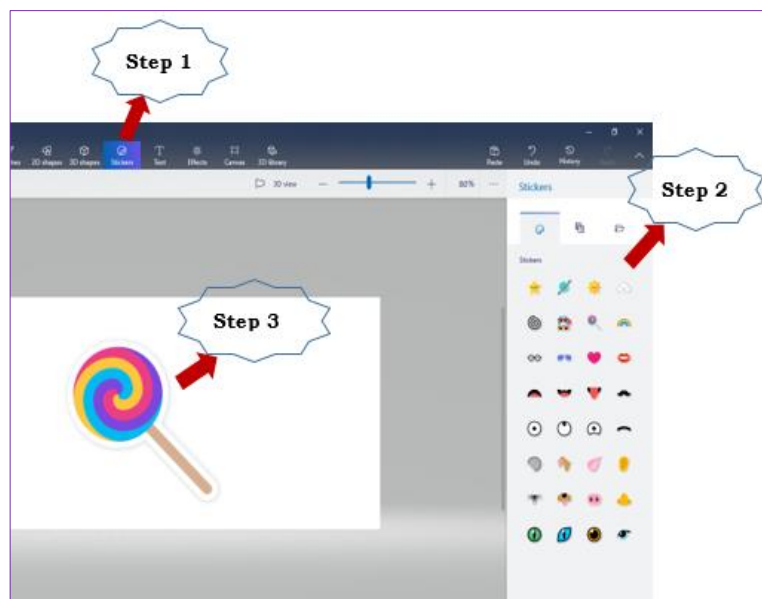
When a 3D object is selected, the side panel provides tools for changing the color and material of the object, editing, and Flipping the object horizontally or vertically.



APPLYING STICKERS

Stickers are like pieces of clip art that can be applied to 3D models. To apply stickers to your drawing follow these steps:

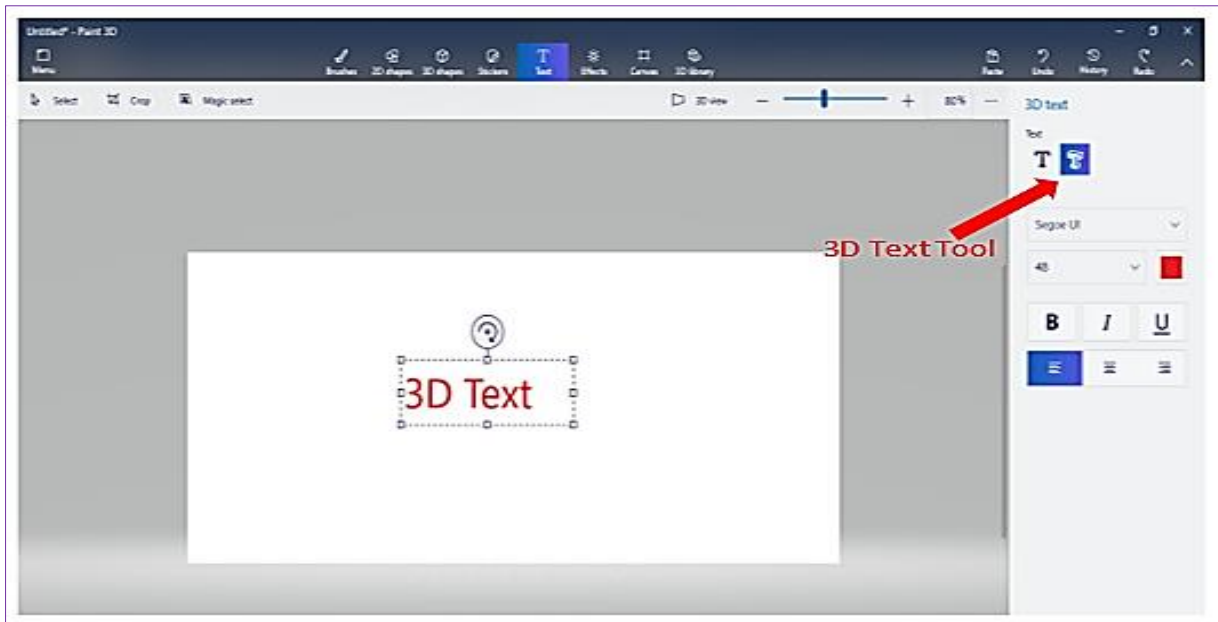
- ✧ Click the Stickers Tool
- ✧ Select a sticker from the Active Tool pane
- ✧ Drag out the sticker on the canvas



CREATING 3D TEXT

The Text tool provides options for 3D text, font, size, color, style, and alignment.

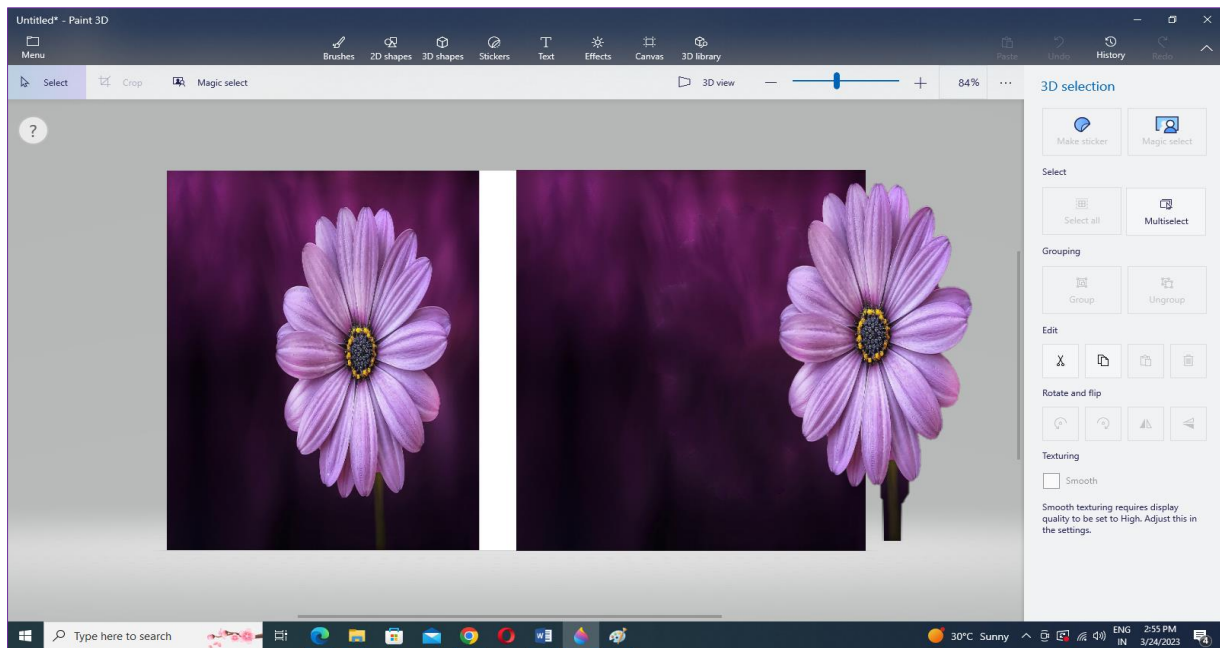
- ✧ Select the Text Tool, then select the 3D Text option
- ✧ Choose your preferred font, font size, color, style, and alignment and then create some text
- ✧ When you are done editing your text click at a point outside of the text box



APPLYING MAGIC SELECT

Magic Select Tool seamlessly removes objects from the background so you can create, design, and edit more easily.

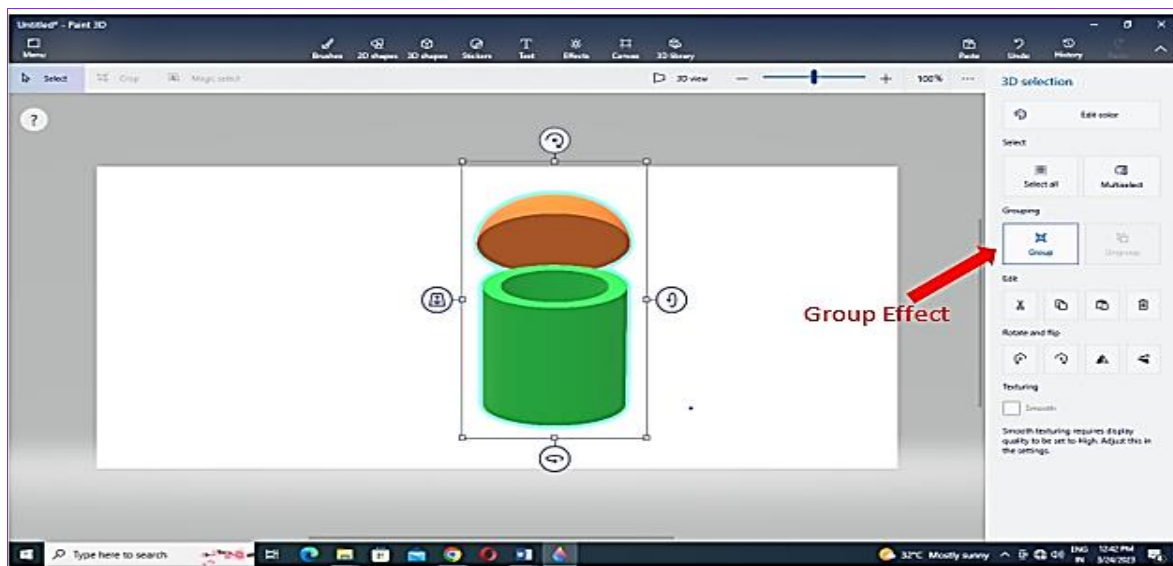
- ✧ Click the Menu icon.
- ✧ Choose the Insert option and select the image you want to remove the background of.
- ✧ Click the Magic Select tool from the upper menu.
- ✧ Drag the corners of the blue box to fit the part you want to keep from the image.
- ✧ Click the Next button.
- ✧ Use the Add and Remove buttons to fine-tune the selection.
- ✧ Click Done when you're finished to have the object cut out and the background removed.



GROUPING EFFECTS

The Grouping Effects is used to group separate objects all together so that you can move, resize and edit them all together. To add group effects, follow these steps:

- ✳ Insert 3D shapes of your choice on the canvas
- ✳ Select a shape, then press Ctrl key and select all other images to be grouped.
- ✳ Select Group option from the Active Tool Pane.
- ✳ Now move, resize and edit the grouped objects all together.



Let's Recapitulate



- ★ MS Paint or Microsoft Paint is an application program included with Microsoft Windows. It can be used for drawing and painting purposes.
- ★ A ribbon holds a set of commands which are arranged in groups. The Home ribbon opens up by default in the Paint program.
- ★ A large white area where you draw and color is called canvas.
- ★ Important tools used in MS-Paint are: Pencil, Eraser, Fill with Color, Color Picker, Text, Magnifier, etc.
- ★ Paint 3D is a free application program from Microsoft to create basic or creative objects by easily combining 2D and 3D tools.
- ★ Rotate Tool in Paint 3D can be used to rotate the shape.
- ★ Paint 3D allows to create creative pictures with 3D Shapes.
- ★ Stickers are like pieces of clip art that can be applied to 3D models.
- ★ The Text tool provides options for 3D text, font, size, color, style, and alignment.
- ★ Magic Select Tool seamlessly removes objects from the background so you can create, design, and edit more easily.
- ★ The Grouping Effects is used to group separate objects all together so that you can move, resize and edit them all together.

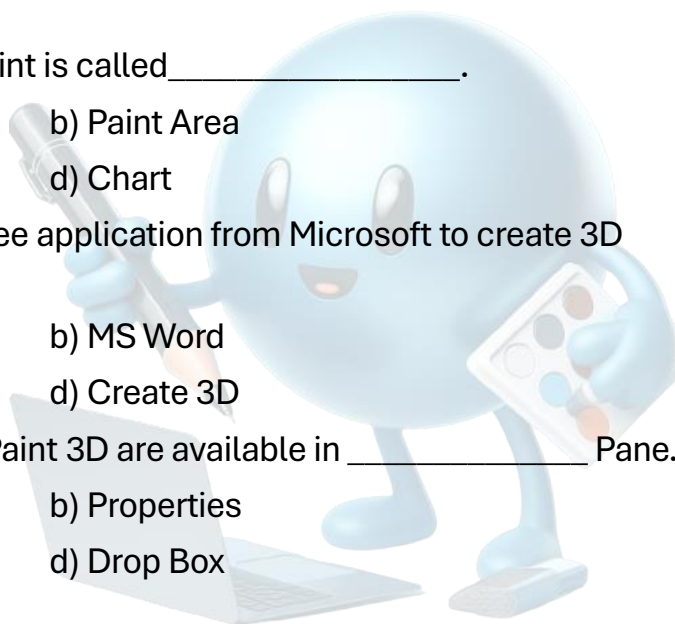
Let's Practice



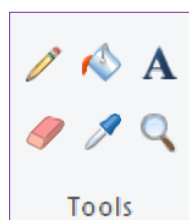
I. Multiple choice questions

1. MS-Paint is the product of _____.
 - a) Adobe
 - b) Microsoft
 - c) Dell
 - d) Apple
2. Which bar is located at the top of Paint Window?
 - a) Status Bar
 - b) Title Bar
 - c) Scroll Bar
 - d) Space Bar

3. The drawing area in Paint is called _____.
 - a) Work area
 - b) Paint Area
 - c) Canvas
 - d) Chart
4. _____ is a free application from Microsoft to create 3D objects.
 - a) MS Paint
 - b) MS Word
 - c) Paint 3D
 - d) Create 3D
5. The selected tools in Paint 3D are available in _____ Pane.
 - a) Active Tool
 - b) Properties
 - c) Passive
 - d) Drop Box



II) Label the following tools



- | | | |
|---|--|---|
| (a)  | (b)  | (c)  |
| (f)  | (e)  | (f)  |

III) Match the following

Column A

1. Color Picker
2. Magnifier
3. Rotate Tool
4. Clone Tool
5. Magic Select Tool

Column B

- This tool removes objects from the background so you can create, design, and edit more easily.
- a) It is used to make a copy of the shape
 - b) Select color from an object and used to color other areas.
 - c) Used to rotate the objects
 - d) Zoom into the parts of the picture or image
 - e)

Let's Do Hands on



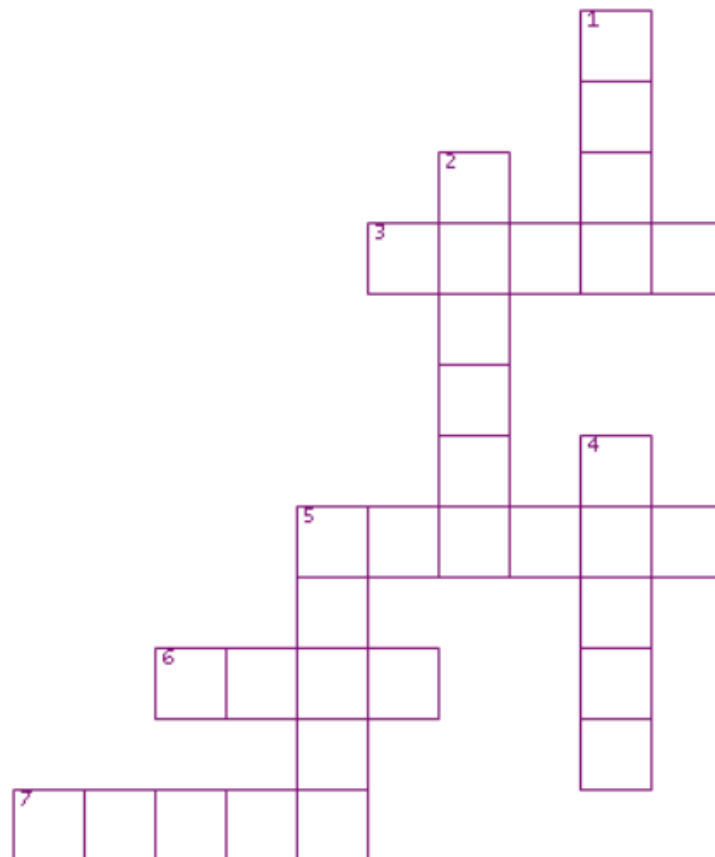
I. Solve the crossword puzzle using the hint(s) given below:

➡ Across

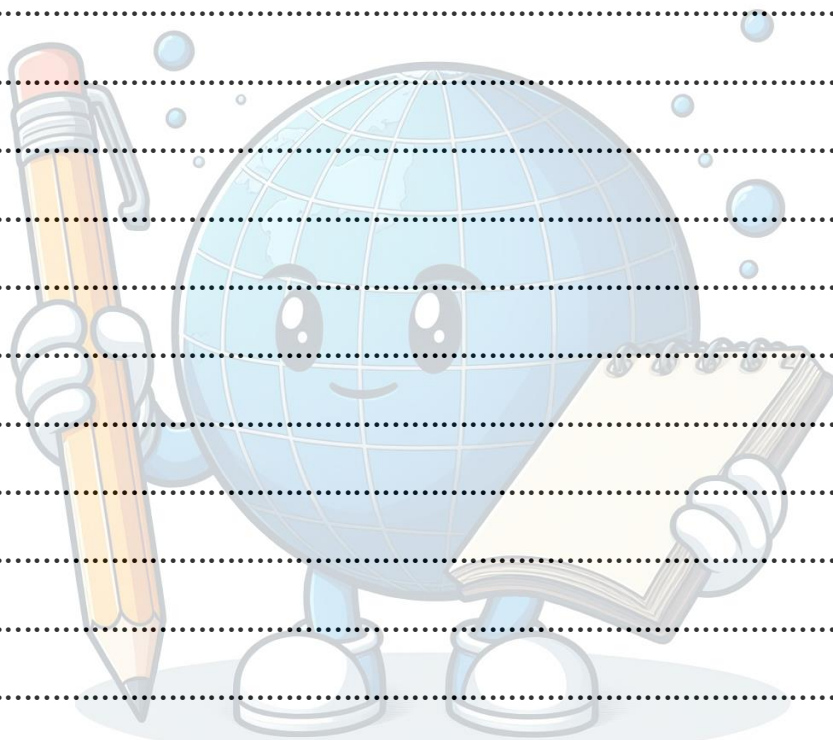
3. Top most bar
5. A large white area in MS Paint
6. This tool removes unwanted areas in the image
7. It's shortcut key is Ctrl+V

⬇ Down

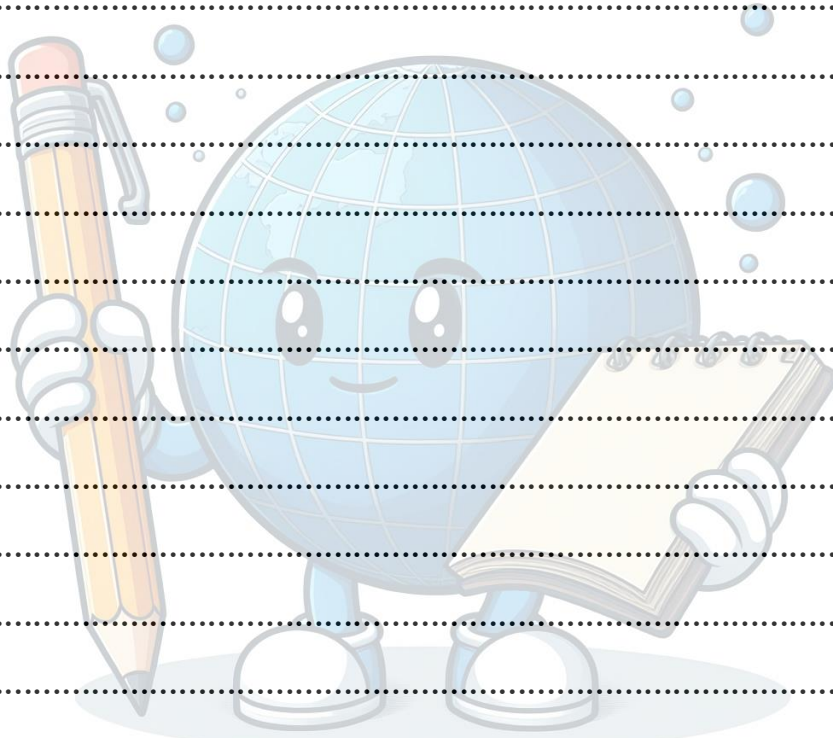
1. This shape can be used to draw circle/ellipse
2. Set of commands arranged in groups
4. A basic graphics editor
5. Tool in Paint 3D to make a copy of shape



Space for Notes



Space for Notes



Answer Key



Let's Practice

I. Multiple choice questions

1. b) 2. b) 3. c) 4. c) 5. a)

II. Label the following tools.

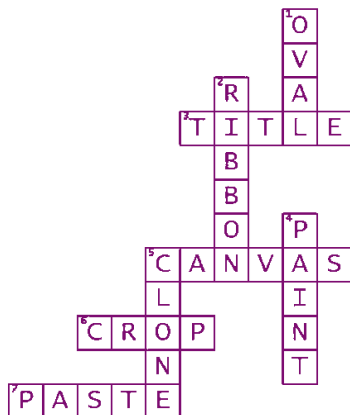
- a) Pencil b) Fill with color c) Text Tool
d) Eraser e) Color Picker f) Magnifier

II. Match the following.

1. c) 2. e) 3. d) 4. b) 5. a)

Let's Do Hands on

I. Crossword Puzzle



INTRODUCTION TO WORD PROCESSING (MS-WORD)

Learning objectives

In this chapter, you will be able to understand:

- ★ Concept of word processing along with its applications
- ★ List various word processing software
- ★ Identify the parts of MS Word Window
- ★ Start, Create, Save and Close a document
- ★ Type text in MS Word
- ★ Apply basic formatting on text
- ★ Change the background



INTRODUCTION

Word processing software simplifies writing tasks by providing tools for editing, formatting, and organizing text. It's a computer program that allows you to write, edit, format, and print documents in a digital environment.

Word processors are used by a wide range of people for various purposes, including:

- ★ Students: Creating essays, reports, and taking notes.
- ★ Professionals: Writing letters, reports, proposals, and presentations.
- ★ Authors: Drafting novels, scripts, and other creative content.
- ★ Anyone who needs to create a document: From grocery lists to business plans, word processors offer a versatile tool for everyone.

Today, there are many word processing applications available, some of the most popular include:

- ★ Microsoft Word
- ★ Google Docs (web-based)
- ★ Apple's Pages (Mac)
- ★ LibreOffice Writer (free and open-source)

A file created in a word processor is known as a **document**.

INTRODUCTION TO MS-WORD

Microsoft word is word processor software developed by Microsoft in 1983. It is the most commonly used word processor software. It is used to create professional quality documents, letters, reports, resumes, etc. MS Word have **.docx** extension. It is a component of the Microsoft Office suite.

START MICROSOFT WORD

Microsoft Word, a core component of the **Microsoft Office suite**, has evolved through numerous versions over the years. For eg. to start MS WORD 2016, follow the given steps:

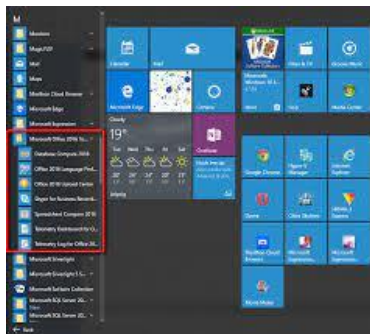
Step 1: Click on the Start button



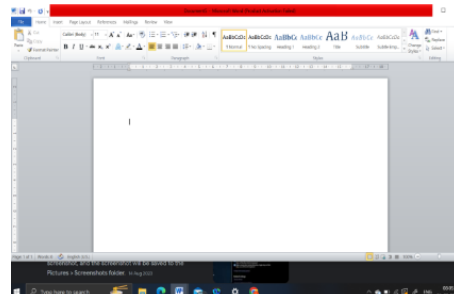
Step 2: Click on the All programs



Step 3: Click on the Microsoft office



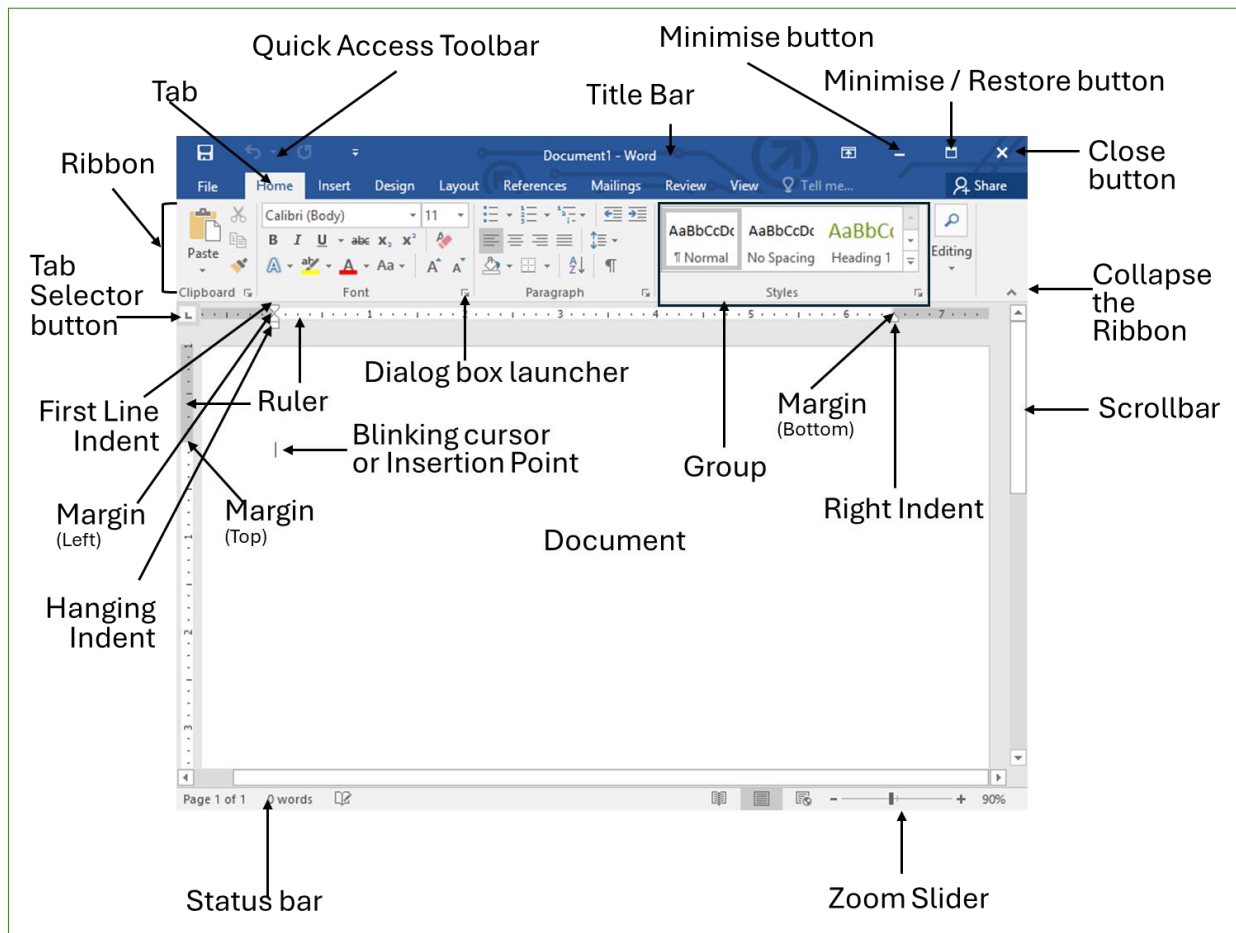
Step 4: Click on the Microsoft office word 2016



PARTS & FUNCTIONS OF MS-WORD WINDOW

The Word screen is divided into various parts – File tab, Quick Access toolbar, Title bar, Ribbon, Ruler bar, Document area, Vertical scroll bar, Horizontal scroll bar and Status bar.

NOTE: Carefully observe the below image and you can find the labelled names, which is indicating the location & function the of Microsoft word.



DO YOU KNOW???

Microsoft Corporation is an American multinational technology company with headquarters in Redmond, Washington.



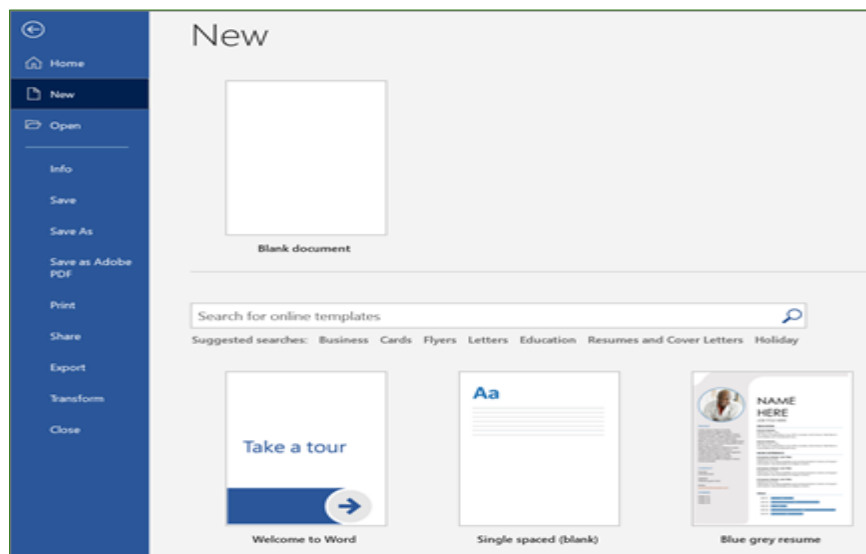
START A NEW DOCUMENT

To create a new document:

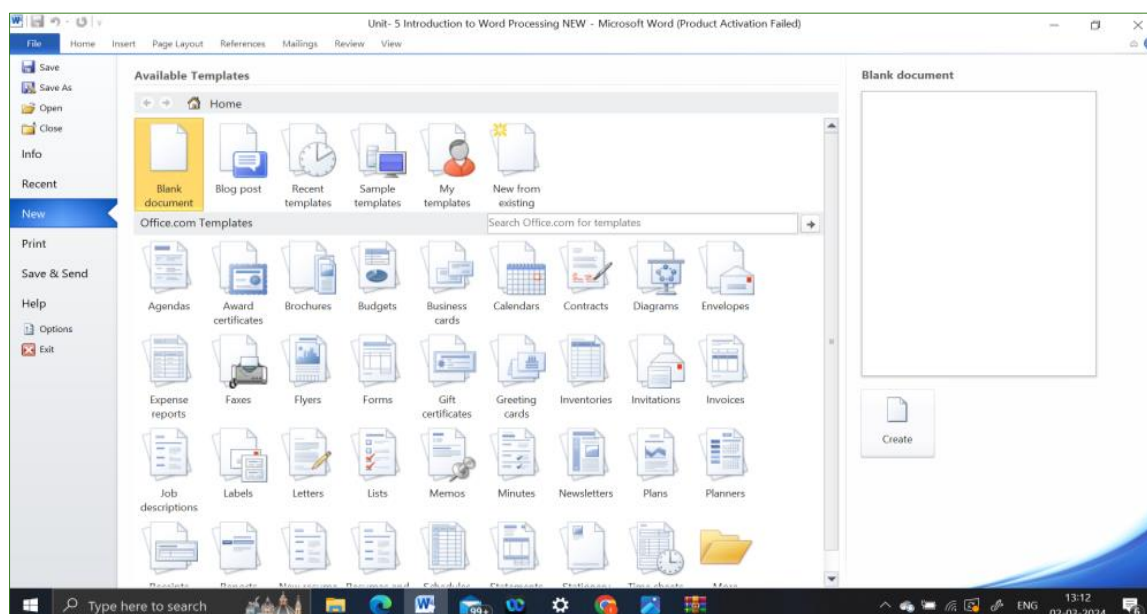
Step 1: Open **Microsoft Word** from start button / menu as stated earlier.



Step 2: Click on the New.



Step 3: Click on the Blank Document





A blank document appears where you can start typing. Word temporarily names new documents in numerical order as Document1, in case of First time, Document 2, Document 3 and so on till you give a new document a file name.

IMPORTANT TABS:

File: The File tab is a section on the office ribbon that gives you access to file functions.

Example:- From file tab, you can access the New, Open, Save, Save As, print, Share , Export , Close, Properties and Recent file option.

Home: The home tab is the default starting point when you open Microsoft office programs and MS Word. It contains essential tools and functions for creating, formation, and managing your documents.

Insert: The insert tab contains several commands for inserting like blank page table, picture, shape, chart, hyperlink, header and footer even page number, WordArt , Equation, Symbols.

Layout: Page layout refers to the overall layout and appearance of your document such as margins, orientation, size, columns, water mark, page color, border etc. to enhance your documents.

References: This tab contains Table Contents, Footnotes, Search, Citation & Bibliography, Caption, Index and Table of Authorities.

Mailings: The mail feature of MS Word allows us to create labels, envelopes, and letters with personalized input.

Review: In review mode, you cannot edit but instead are allowed to add suggestions to the document in the form of comments or tracked changes. Mainly its spelling & research and translate etc.

View: The view tab enables you to switch between Normal or Master Page, and Single Page (or) Two Page Spread views. This tab also gives you control over showing boundaries, guides, rulers, and other layout tools, zooming the size of your view of the publication, and managing Publisher windows you have open.

Help: This tab contains Help, Contact Supports, Feedback and Show Training.

The **Home tab** will open by default and it has **five segments**.

Clipboard: It has Paste, Cut, Copy and Format Painter Options.

Font: It has Fonts, Size, Bold, Italic, Underline, Strikethrough, Subscript, Superscript, Increasing Font size, Decrease Font size, Change Case, Clear All formatting, Text Effects and Typography, Text Highlight Color and Font Color.

Paragraph: It has Bullets, Numbering, Multilevel List, Decrease Indent, Increase Indent, Sort, Show/Hide Paragraph Marks, Align Left, Centre, Align Right, Justify, Line & Paragraph Spacing, Shading and Borders.

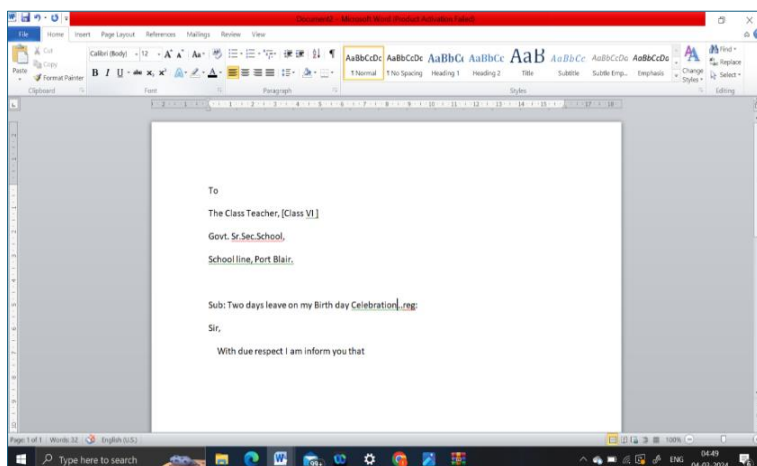
Styles: A style is a set of formatting like Normal, No space, Heading 1, Heading 2, Title, Subtitle, and Emphasis etc.

Editing: The Editing is the process of making changes or corrections in a document. It includes alterations to the text itself, moving or copying items to other locations and applying formatting option to the document itself and items within it.

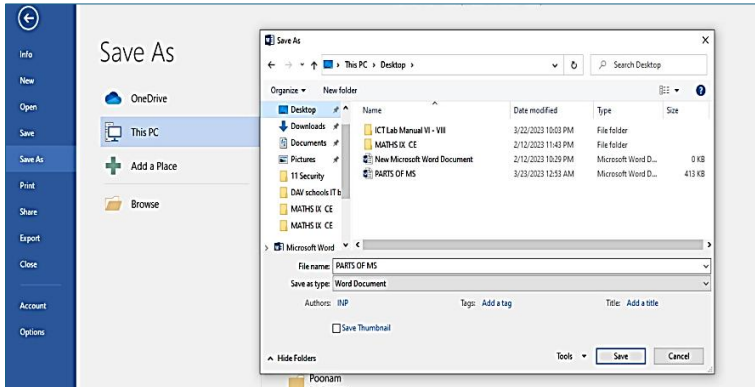
SAVING DOCUMENTS

It is required First time after created /finishing or modified the text documents. Follow the given steps.

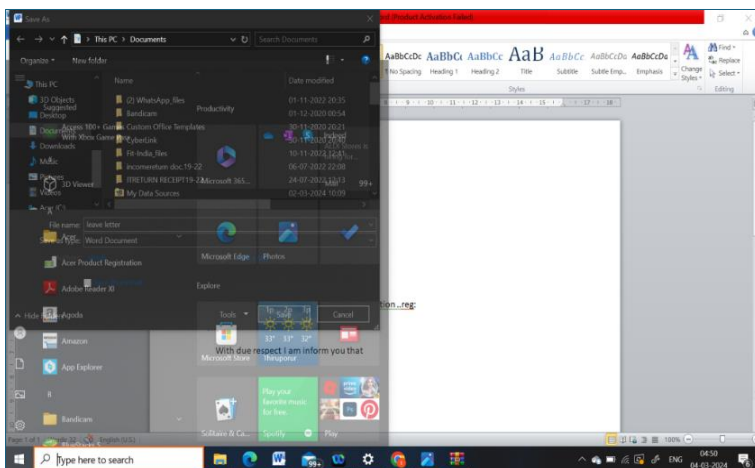
Step 1: Click on **File**.



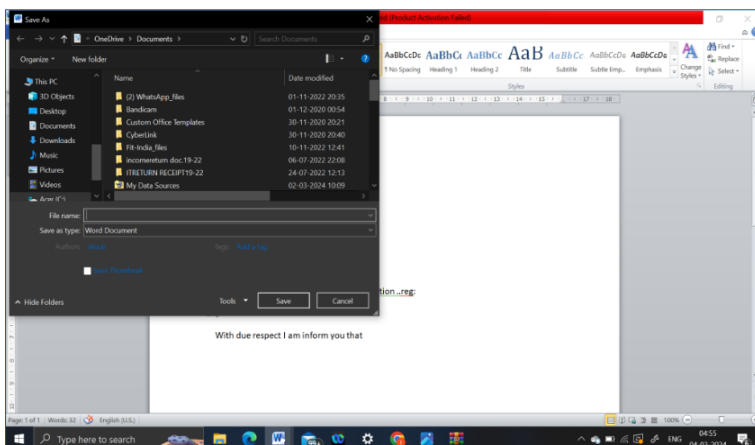
Step 2: Click on **Save As**.



Step 3: Save as window will be appears on computer screen.



Step 4: Click on the **File name** option and Assign or Type a file name for your document.



Step 5: Click on the **Save** Button.

Note: You must save every document that you have created otherwise, if the computer is switched off or power off by mistake, you will lose your document. The Save commands saves the current document.

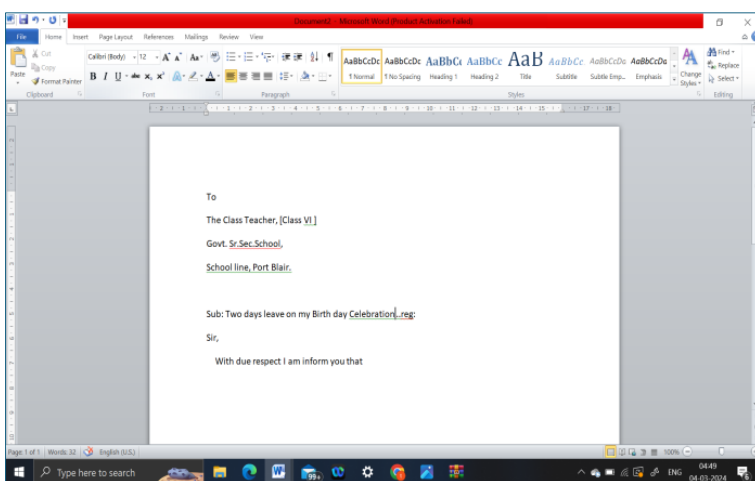
Word automatically adds a **.DOCX** extension.

OPENING A DOCUMENT

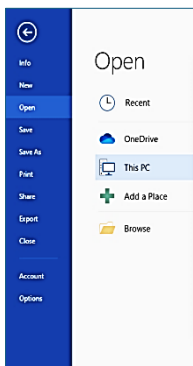
The open command opens an existing document.

To open a document that you have already created and saved:

Step 1: Click on File option in MS Word

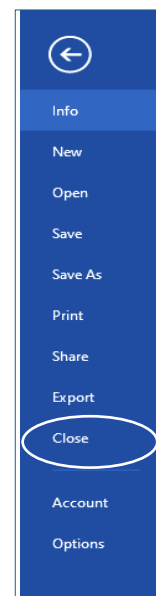
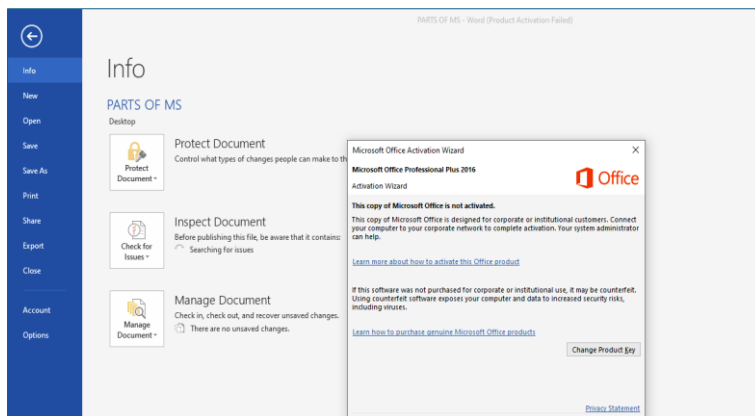


Step 2: Click on Open option in MS Word



The Open dialog box is displayed. Your screen will look like this:

Step 3: Click on File name option and type the Existing file name, click on the open button in the dialog box.



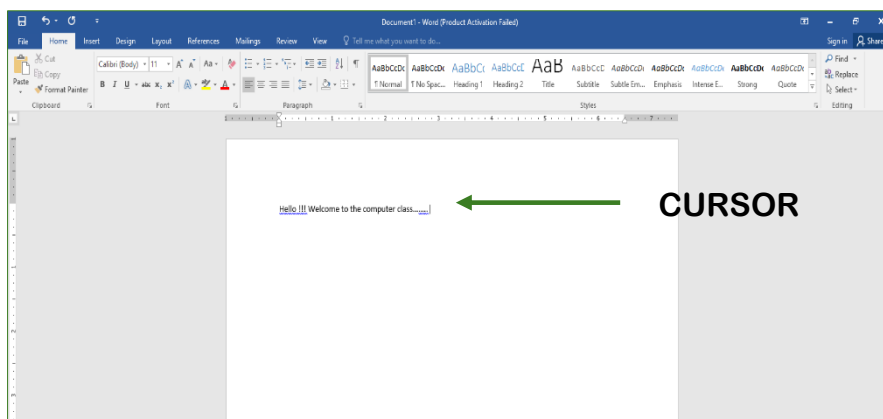
CLOSING A DOCUMENT

We must close a document after we finish working with it. The Close command closes the document window.

To close a document click: File option > Click on Close

TYPING TEXT IN A WORD DOCUMENT

When you open the Microsoft Word you will see a blinking cursor or insertion point in the text area below the ribbon. Now, as you start typing, the typed words will appear on the screen in the text area.



SELECTING A TEXT

Steps to select the text are listed below.

- ★ Bring the cursor to the place from where you want to select the text.
- ★ Click and drag the mouse button till the end point of your selection.
- ★ Release the mouse button when the text is selected.

The highlighted text is the selected text. We can make changes in it as desired. After editing, we can deselect the text by clicking anywhere outside the selection area on the page or by pressing an arrow key on the keyboard.

Selection with mouse

The following table shows the techniques for selection of text with the mouse:

Selection	Technique
A word	Double-click on the word.
Several words or lines	Drag the mouse over the words, or hold down the Shift key while using the arrow keys.
Entire document	Press Ctrl + A or use the mouse to select all the text

Selection with Keyboard

Selecting the text with the help of the keyboard (Shortcut Key)

- ★ **One word or group of words:** Place the cursor at one end of the word. Then, press Shift and move the cursor with the help of the arrow key to the other end of the word.
- ★ **A group of lines:** After selecting the line, keep pressing the Down arrow key along with the Shift key.
- ★ **Entire document:** Select the entire document by pressing Ctrl + A.

DO YOU KNOW???

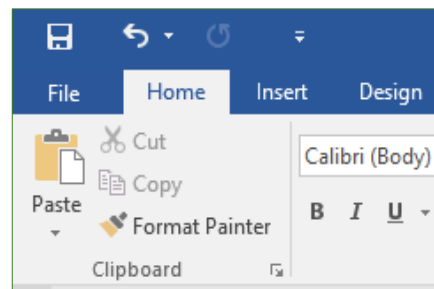
The shortcut to exit MS-Word is **Alt+F4**.



CUT, COPY AND PASTE COMMANDS

The Cut, Copy and Paste commands are in the Clipboard group of the Home tab.

The **Cut** command removes the selected text from its place, while the **Copy** command creates a duplicate of the selected text without changing the original text in any way. We can use the **Paste** command to place this cut or copied text where you want.



The various steps to cut/copy the text are:

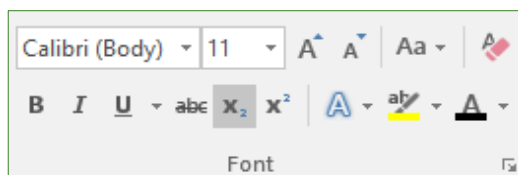
- ★ Select the text that has to be cut / copied.
- ★ Click on the Cut/Copy button from the 'Clipboard' group of the 'Home' tab.
- ★ Take the cursor to the spot where the text has to be pasted. Right-click the mouse button and click on the Paste button.

Keyboard Shortcut

Cut → Ctrl + X,
Copy → Ctrl + C
Paste → Ctrl + V

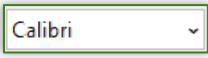
BASIC FORMATTING OF TEXT

Formatting means changing the appearance of the document to make it look attractive. This can be done by changing the font type, font size, font color, bold facing, italicizing, underlining the text, etc.



The Font group of the Home tab has many options to format the text in a document.

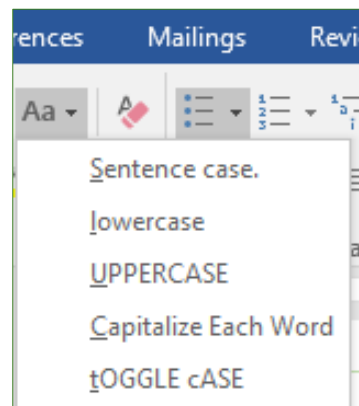
Option	Description	Shortcut
	Bold: Converts the text to bold & vice versa.	Ctrl + B
	Italics: Italicizes the text and vice versa.	Ctrl + I
	Underline: Underlines the text & vice versa.	Ctrl + U

Option	Description	Shortcut
	Font: Changes the font type.	
	Font Size: Changes the font size.	Ctrl +] to increase Ctrl + [to decrease
	Changes the text color.	

CHANGE CASE

Word changes the case of the selected text. These are the options:

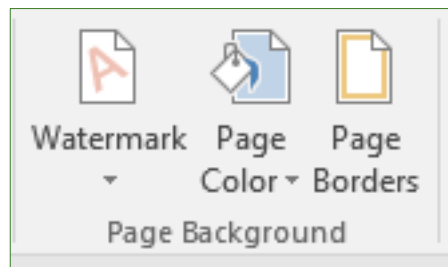
- ★ **Sentence Case:** It capitalizes the first letter of each sentence in the selection.
- ★ **Lowercase:** It changes all the capital letters to small letters.
- ★ **Uppercase:** It changes all the small letters to capital letters.
- ★ **Capitalize each word:** It changes the first letter of every word to capital letter.
- ★ **Toggle Case:** It changes all the capital letters to small letters and vice versa.



CHANGING PAGE BACKGROUND

Word allows you to add a background (color, gradient, pattern, texture, picture) to a whole document.

You can do this by going to the 'Design' tab on the ribbon, choosing 'Page Color' under the 'Page Background' section and choosing a color.



Let's Recapitulate



- ★ MS Word is the most commonly used word processor and is a part of MS Office package.
- ★ The documents created in MS Word have .docx file extension.
- ★ MS Word has tabs like File, Home, Insert, Layout, References, Mailings, Review, View.
- ★ The Cut, Copy & Paste commands are part of the Clipboard group in Home tab.
- ★ Basic formatting of text includes font type, font size, bold, italic, underline, font color etc. which are part of the Font group in Home tab.



Let's Practice

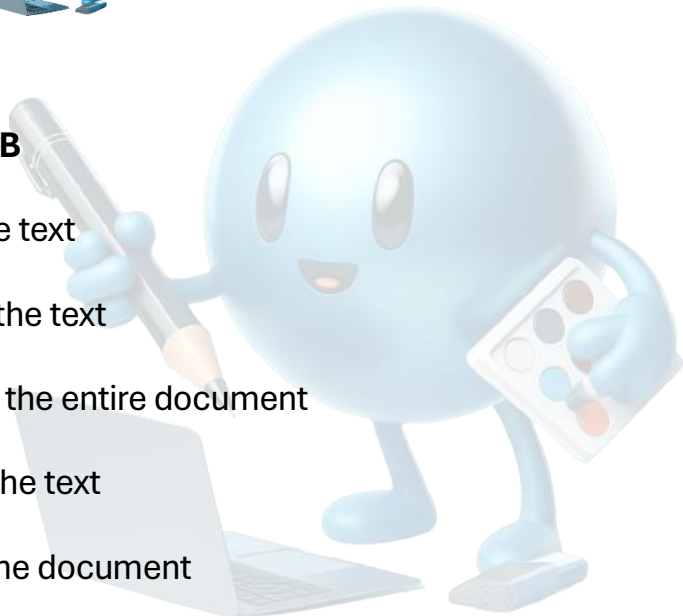
I. Match the following

Column A

1. Ctrl + S
2. Ctrl + C
3. Ctrl + V
4. Ctrl + X
5. Ctrl + A

Column B

- a) Cut the text
- b) Paste the text
- c) Select the entire document
- d) Copy the text
- e) Save the document



- II. In the box, write all the letter keys in the same order you find on the keyboard.

- III. Tick the correct answers in the box.

- The file extension of a word document is ?
☐ *.pptx ☐ *.bmp ☐ *.docx ☐ *.pdf
- Shortcut to select all the text in a document ?
☐ Ctrl +S ☐ Ctrl + A ☐ Ctrl + X ☐ Ctrl + V

- IV. Say true or false

- To edit a text in a document, it has to be first selected.
- Ctrl + A makes us to select the entire text in a document.
- We cannot select the text with the help of the mouse.
- We use the delete key to remove the letters on the left of the cursor.

Let's Do Hands on



- I. Word documents project

At least one word document should be created by each student.

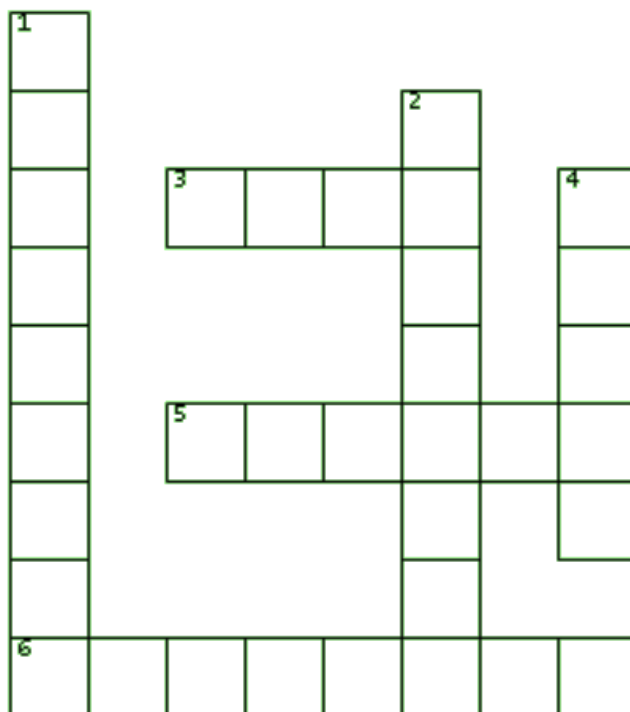
II. Crossword puzzle

➡ Across

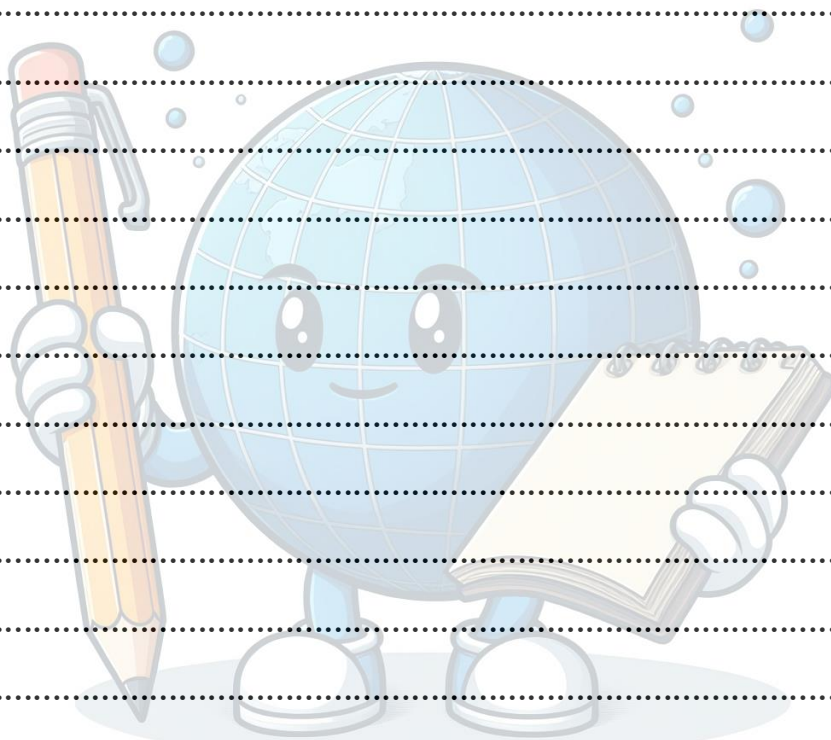
3. This tab is default tab in MS Word
5. WordArt is available in this tab
6. File created in word processor

⬇ Down

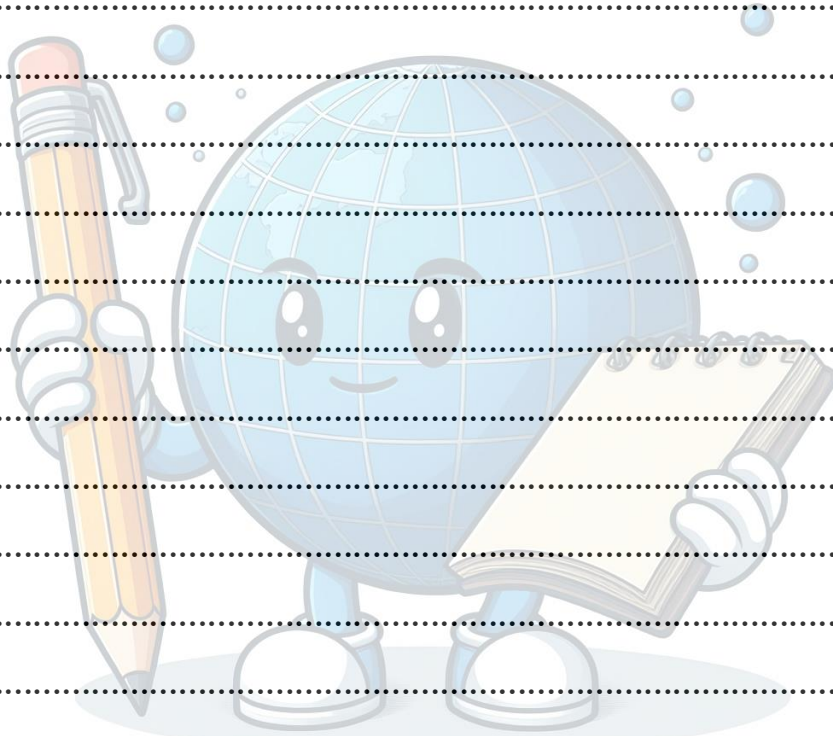
1. Here data like text & pictures are stored temporarily
2. This case capitalizes the first letter of each sentence
4. Ctrl+V is its keyboard shortcut



Space for Notes



Space for Notes



Answer Key



Let's Practice

I. Match the following

1. e) 2. d) 3. b) 4. a) 5. c)

II. All the letter keys available on keyboard



III. Tick the correct answers in the box

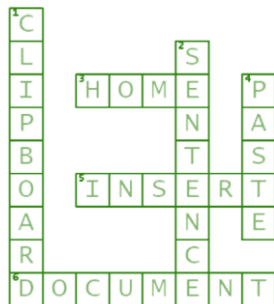
1. *.docx 2. Ctrl + A

IV. Say true or false)

1. (T) 2. (T) 3. (F) 4. (F)

Let's Do Hands on

I. Crossword Puzzle



COMPUTER NETWORKS

Learning objectives

In this chapter, you will be able to understand:

- ★ Definition
- ★ Types of network
- ★ Internet
- ★ Different Terminology related to networks
- ★ Uses of networking
- ★ Connecting to internet
- ★ WWW concepts



COMPUTER NETWORK

A **Computer Network** is an interconnection among two or more computers. Such interconnection allows computers to share data and resources among each other. The network size may vary from small to large depending on the number of computers it connects. A computer network can include different types of hosts (also called nodes) like server, desktop, laptop, cellular phones etc.,



Types of Networks

Classification of Computer Networks:

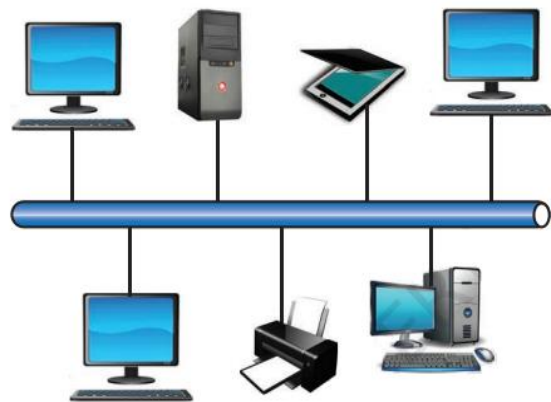
Personal Area Network (PAN)

- ★ It is the smallest network of computers.
- ★ Bluetooth devices could be used to connect devices.
- ★ It has a 10-metre range of connectivity.
- ★ PAN network enables person's personal devices to connect with each other



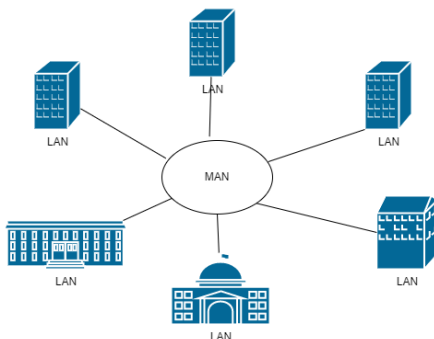
Local Area Network (LAN)

- ★ LAN connectivity is within a building like School, College, Office, etc.
- ★ The distance range is from 10mtr. to 100 mtr.
- ★ **Data transfer is quick.**



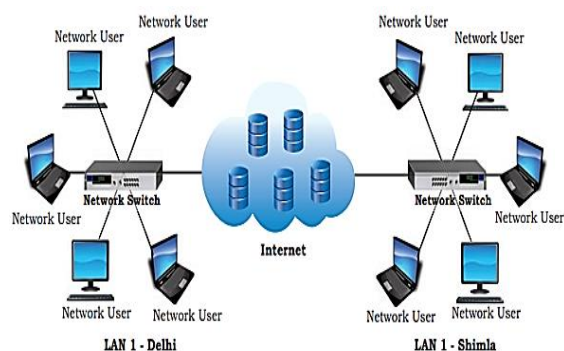
Metropolitan Area Network (MAN)

- ★ MAN connectivity is within a city like Delhi, Chennai, etc.
- ★ The distance range is from 5kms to 50kms.
- ★ Example- Cable TV Connection.
- ★ **Data transfer is average.**



Wide Area Network (WAN)

- ★ WAN connectivity is within a states, countries, and continents.
- ★ The distance range is from 100 or more kilometers.
- ★ Example-Internet.
- ★ **Data transfer is low to average.**



Advantages of networks

- ★ Used to share resources (**Printer, Files, Memory etc.**)
- ★ Internet is widely used for communication.
- ★ Connectivity- It helps people to get in touch from anywhere.

INTRANET & INTERNET

Intranet

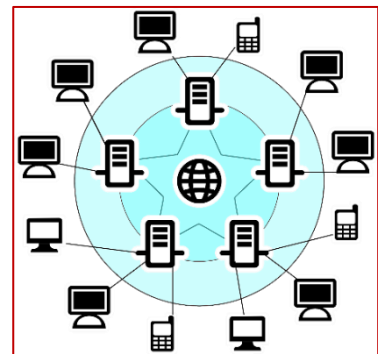
The intranet is a computer network for sharing information within an organization itself. Like network in Multi-National Companies(MNCs).

Internet

The Internet is a global system of interconnected computer networks. It helps people to communicate, share information and access lots of different things like websites, videos and games.

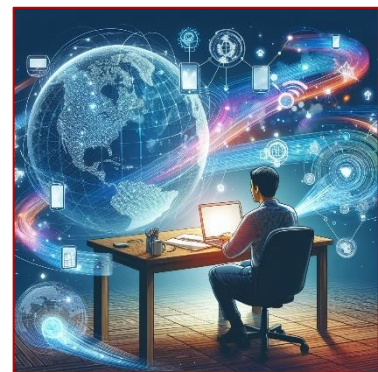


The internet's greatest strength lies in its accessibility. By using a mobile, computer or any other internet-enabled device you can connect from virtually anywhere in the world.



Advantages of the internet

- ★ E-mail- sending and receiving mails (e.g. Gmail, Microsoft outlook)
- ★ Huge amount of information available on the internet.
- ★ Instant Messaging (IM)- sending and receiving quick text messages instantly. (e.g. WhatsApp, Messenger, Skype, Snapchat)
- ★ E-commerce- (e.g. Amazon, eBay, Flipkart)
- ★ Entertainment- (e.g. Amazon prime, Netflix, Jio Hotstar)



WWW, WEB BROWSERS & SEARCH ENGINES

World Wide Web (WWW)

World Wide Web is a system of interlinked hypertext documents accessed via the Internet. With a web browser, one can view web pages that may contain text, images, videos, and other multimedia, and navigate between them via hyperlinks.



Web Browsers

A Web Browser is a software used to view websites and acts as an interface between the user and the World Wide Web. A Web server is a computer that stores web sites and their related files for viewing on the Internet.

Some of web browsers include **Google Chrome, Safari, Mozilla Firefox, Opera, Edge** etc.



Search Engines

A search engine is a service that allows Internet users to search for content via the World Wide Web (WWW). A user enters keywords or key phrases into a search engine and receives a list of Web content results in the form of websites, images, videos, etc. Ex: Google, Yahoo, Bing etc.,



DO YOU KNOW???

The Web was invented by English computer scientist **Tim Berners-Lee**.



Let's Recapitulate



- ★ Computer network is an interconnection between one or more computer.
- ★ Types of networks PAN, LAN, MAN and WAN.
- ★ Intranet is a computer network for sharing information within an organization
- ★ Internet is a global system of interconnected computer networks that use the standard Internet protocol suite to serve billions of users worldwide.
- ★ E-commerce is used to buy and sell goods electronically over the internet
- ★ WWW is also known as web. World Wide Web is a system of interlinked hypertext documents accessed via the Internet.
- ★ Web browser is a software used view websites.
- ★ Search engine is a service that allows Internet users to search for content.

Let's Practice



I. Fill in the blanks

1. is an interconnection between one or more computers.
2. enables personal devices to connect with each other.
3. is a software used to view websites.
4. The abbreviation of WWW is
5. is used to buy and sell goods electronically over the internet.

II. Short answer type questions

1. Define Computer networks?
2. Give any three advantages of computer networks.
3. Compare LAN and WAN.

4. Write the advantages of internet.
5. Define:
 - a) WWW
 - b) Web Browser
 - c) Search Engine

III. Match the followings

Column A

1. LAN
2. WWW
3. MAN
4. E-mail
5. PAN
6. WAN

Column B

- a) Personal Area Network
- b) Wide Area Network
- c) Electronic Mail
- d) World Wide Web
- e) Local Area Network
- f) Metropolitan Area Network

IV. Identify the icons



Let's Do Hands on



I. Lab Activity

Identify the components of computer network which is available in your computer lab. List out any five components.

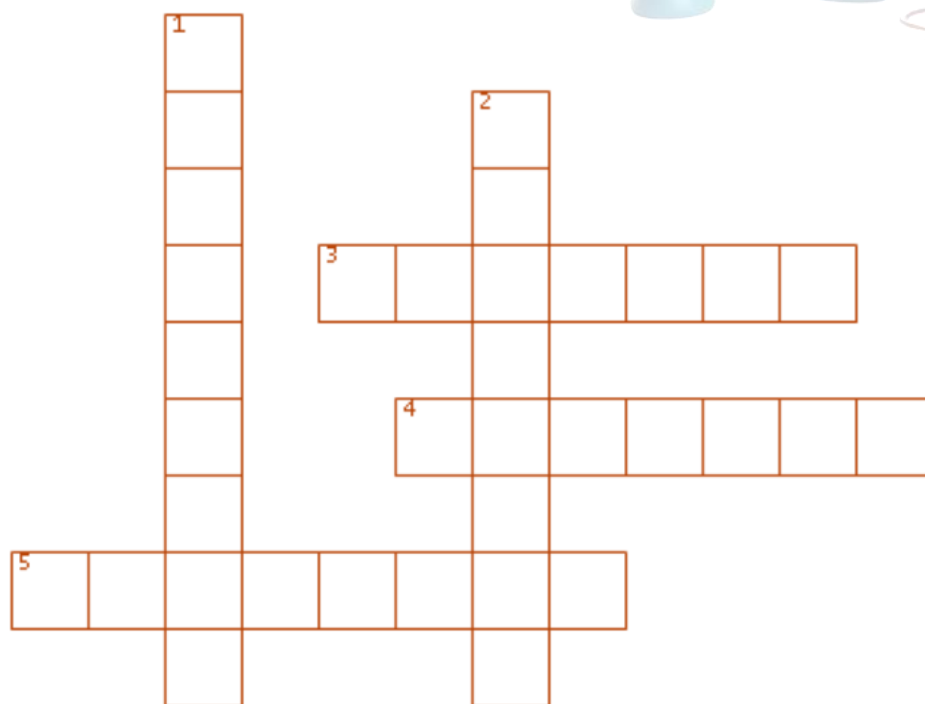
II. Crossword Puzzle

➡ Across

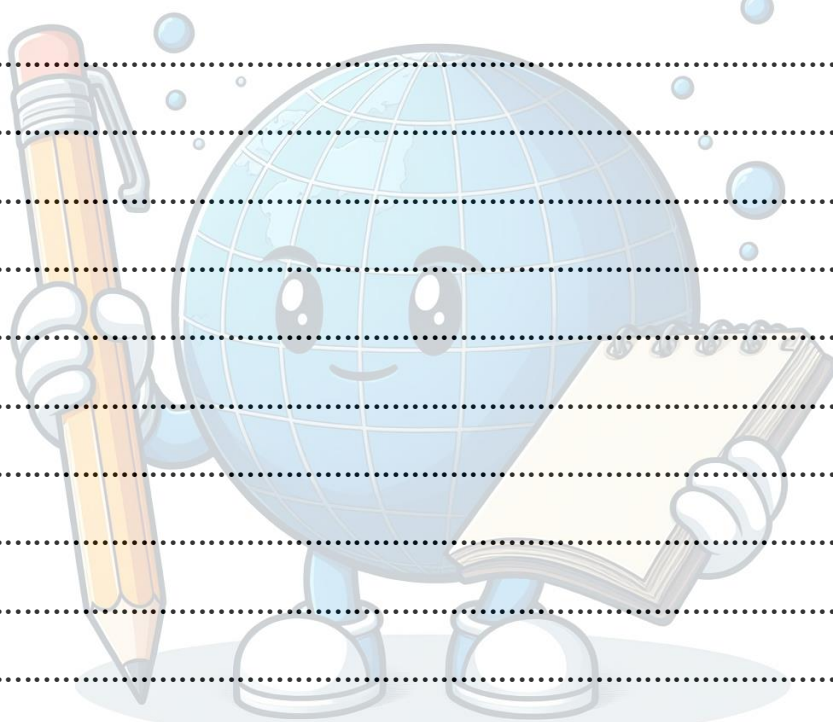
3. Interconnection among two or more computers
4. Google chrome is a web _____
5. Network for sharing information within an organization

⬇ Down

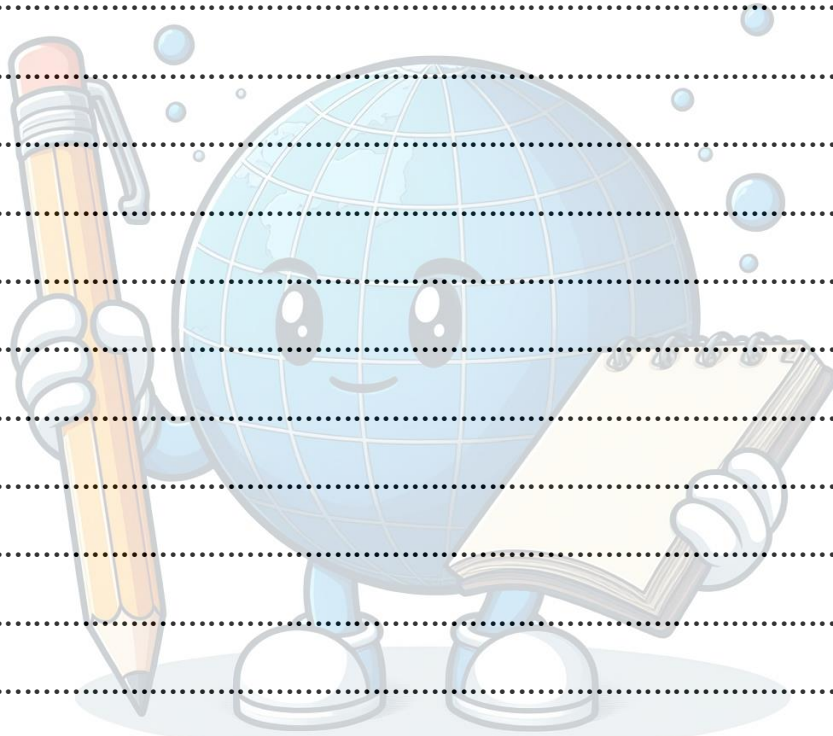
1. These devices could be used in PAN
2. Global system of interconnected computer networks



Space for Notes



Space for Notes



Answer Key



Let's Practice

I. Fill in the blanks

- | | | |
|---------------------|---------------|----------------|
| 1. Computer network | 2. PAN | 3. Web browser |
| 4. World Wide Web | 5. E-commerce | |

III. Match the followings

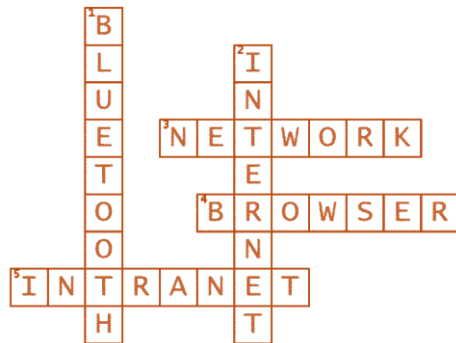
- | | | |
|-------|-------|-------|
| 1. e) | 2. d) | 3. f) |
| 4. c) | 5. a) | 6. b) |

IV. Identify the icons

- | | | | |
|--------------------|------------------|----------------------|----------|
| 1. Mozilla Firefox | 2. Google Chrome | 3. Internet Explorer | 4. Opera |
|--------------------|------------------|----------------------|----------|

Let's Do Hands on

I. Crossword Puzzle



CYBER ETHICS

Learning objectives

In this chapter, you will be able to understand:

- ★ What is Digital Foot Print
- ★ Online Safety & Security
- ★ How to stay safe online
- ★ Safe use of Internet
- ★ Cyber threat viz. Phishing & Spoofing
- ★ Cyber protection from frauds & scams



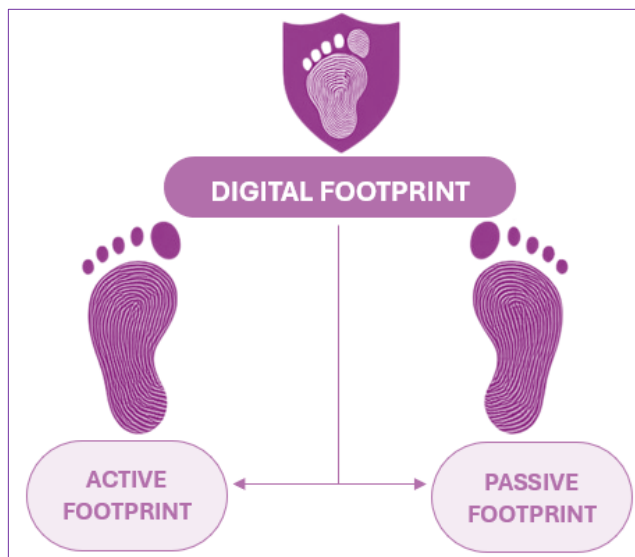
DIGITAL FOOTPRINTS

A digital footprint is data that is left behind when users have been online.



There are two types of digital footprints which are **passive** and **active**.

- ✧ A **Passive Footprint** is made when information is collected from the user without the person knowing this is happening.
- ✧ An **Active Footprint** is where the user has deliberately shared information about themselves either by using social media sites or by using websites.



ONLINE SAFETY AND SECURITY

Online Safety is being aware of the nature of the possible threats that you could encounter while engaging in activity through the Internet, these could be

- ✧ Security threats
- ✧ Protecting and managing your personal data
- ✧ Online reputation management,
- ✧ Avoiding harmful or illegal content.



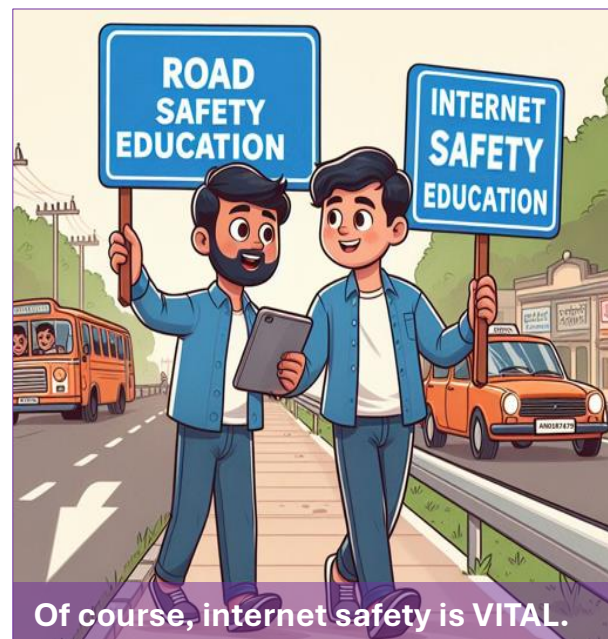
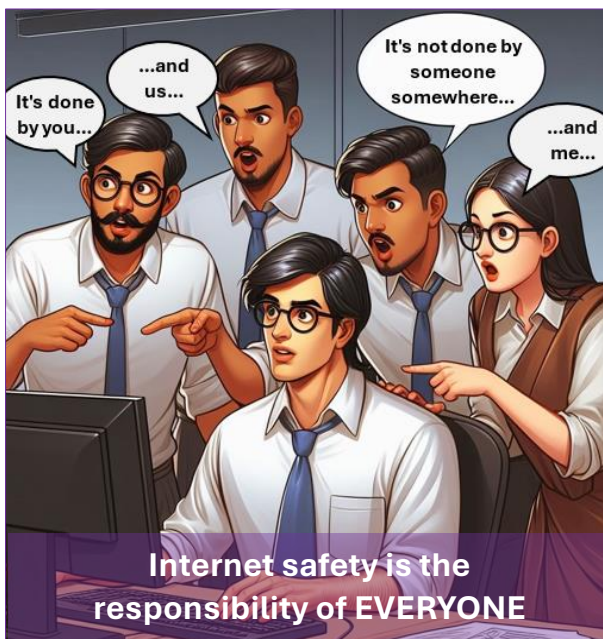
To stay safe online

- ✧ Use unique passwords. Using the same password for logging in to multiple accounts might seem easy and more convenient.
- ✧ Log out from services. ...
- ✧ Keep your software up to date.
- ✧ Beware phishing attacks.
- ✧ Take back-ups.



SAFE USE OF INTERNET

- ★ Never click on a doubtful link on a web page.
- ★ Never send through a mail for which you are not sure about its genuineness.
- ★ Make sure your passwords are strong and are changed frequently.
- ★ Weak passwords can be easily cracked.
- ★ Do not set passwords as names, mobile numbers, or date of birth.
- ★ Never share your personal information such as date of birth, home address, personal phone number, account details, passwords, credit and debit card details, work history details.



CYBER THREAT AND PROTECTION

Cyber Threat

Cyber threats such as Phishing, email spoofing, chat spoofing, etc. can increase the chances of users getting compromised.

Phishing

Phishing is stealing personal information through fraud mails.



Spoofing

Spoofing is hiding a communication from an unknown persons or sources as being from a known, trusted source.



Cyber Protection

Cyber Protection is the practice of protecting computers, servers, mobile devices, electronic systems, networks, and data from any malicious attacks. Cyber protection uses tools and techniques to protecting from threats, data and the systems that store and process it against attacks that could result in unauthorized access to and disclosure of that data.

We can protect ourselves from these threats & attacks by knowing them in details.

Email fraud

An email asking you to change your password for updating your account by clicking on a link.



Lottery fraud

A text message on your phone informing you that you have won a big sum of money.



Greeting Card Fraud

An email asking you to download a card from a secret admirer.



Gaming scam

A pop-up in your online game offering 100 free game coins if you follow the link and watch a 2-minute video.



Messaging fraud

A message telling you that security has been breached on your messaging app and that you need to click on a link to reset your password.



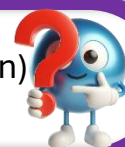
Charity scam

A pop-up on your screen asking you to donate to those affected by recent floods.



DO YOU KNOW???

“**Phishing**” is a homophone (same pronunciation) derived from the word “fishing.”



Let's Recapitulate



- ★ A digital footprint is the track of a person's online activities.
- ★ Internet users create their digital footprint either actively or passively.
- ★ We should be aware of the nature of possible online threats and use internet safely.
- ★ Cyber threats such as phishing and spoofing increase the chances of users getting compromised, whereas cyber protection uses tools & techniques to protect from these threats & attacks.
- ★ The Cyber Safety helpline number in India is 1930. You can call this number to report cybercrime or online frauds.



Let's Practice

I. Multiple choice questions

1. What is one difference between online and face-to-face audio communication ?
 - a) Body language is more important in online communication
 - b) People tend to be more honest online
 - c) Miscommunications happen more frequently when interacting face-to-face
 - d) You can't tell a person's tone of voice when interacting online
2. Internet trolls aim to...

a) Start arguments	b) Make friends
c) Quietly observe	d) Lead discussions

3. Why might people act meaner online than in person?
 - a) Typing gives them time to consider what to say
 - b) They can stay anonymous
 - c) Most people have fewer friends online
 - d) Social media sites encourage users to argue
4. Posting something on a social networking app is most like...
 - a) Handing in a homework assignment
 - b) Saying something to your family over dinner
 - c) Speaking to a friend on the phone
 - d) Announcing something in a public place
5. The best way to respond to trolls is to..... them.
 - a) Threat
 - b) Ignore
 - c) Copy
 - d) Insult
6. Which of these is the best text to tell your mother that you'll be an hour late for dinner?
 - a) b there in 60, cu l8r!
 - b) I'll be home in an hour. See you later!
 - c) BE THERE IN AN HOUR. SEE YOU LATER!
 - d) I'm running late. I'll be there in an hour



II. Read through each of the following situations.

Then, write down whether you think it is a **good idea** or a **bad idea**.

1

I will not accept a friend request from someone I don't know.....

2

I will share my school timetable on my social media account. Everyone will be really impressed, knowing how hard I work.....

3

When my new online friend asks to meet me at the coffee shop, I inform my parents.....

4

When I go to the mall, I should always keep my phone's locator turned on so that my family can find me easily.....

5

I love filling out those silly surveys online that ask me questions like the name of my school or my first pet's name.....

6

I just received an email saying that I've won a trip to Paris! I am so excited. I click on the link to see what sites we would visit.....

7

I just changed my email password and it is a strong one! I included numbers, capital letters, and symbols. It's good and I am never going to tell anyone what it is.....

Let's Do Hands on



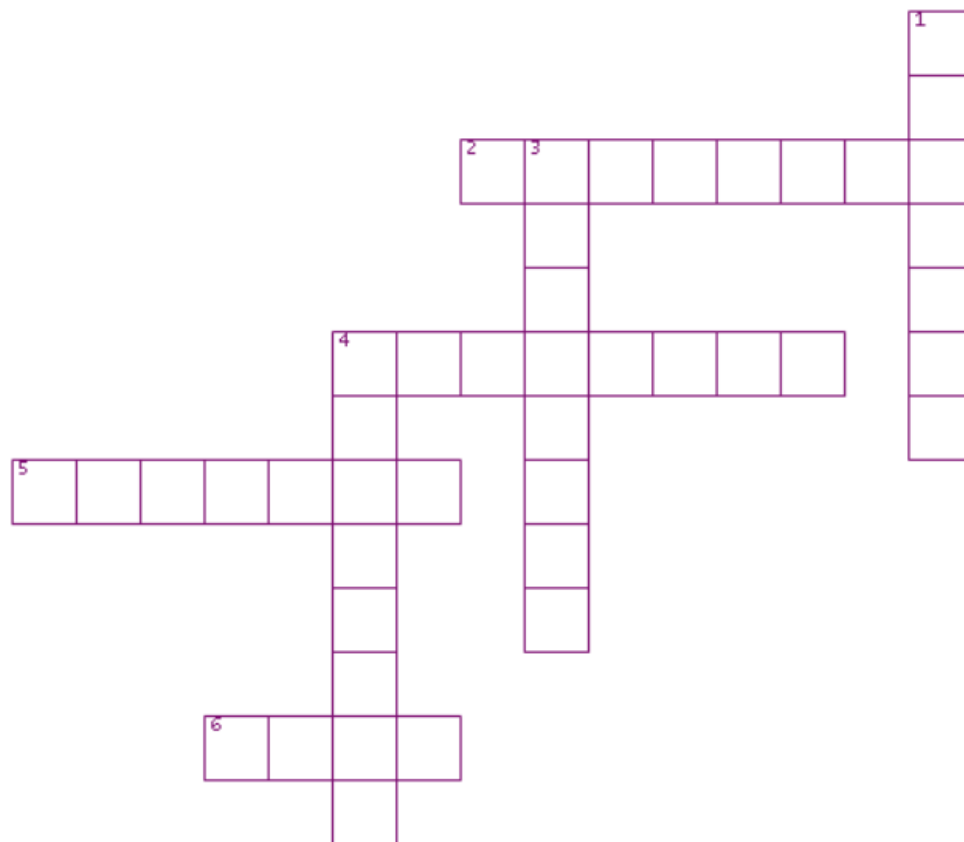
I. Solve the crossword puzzle using the hint(s) given below:

➡ Across

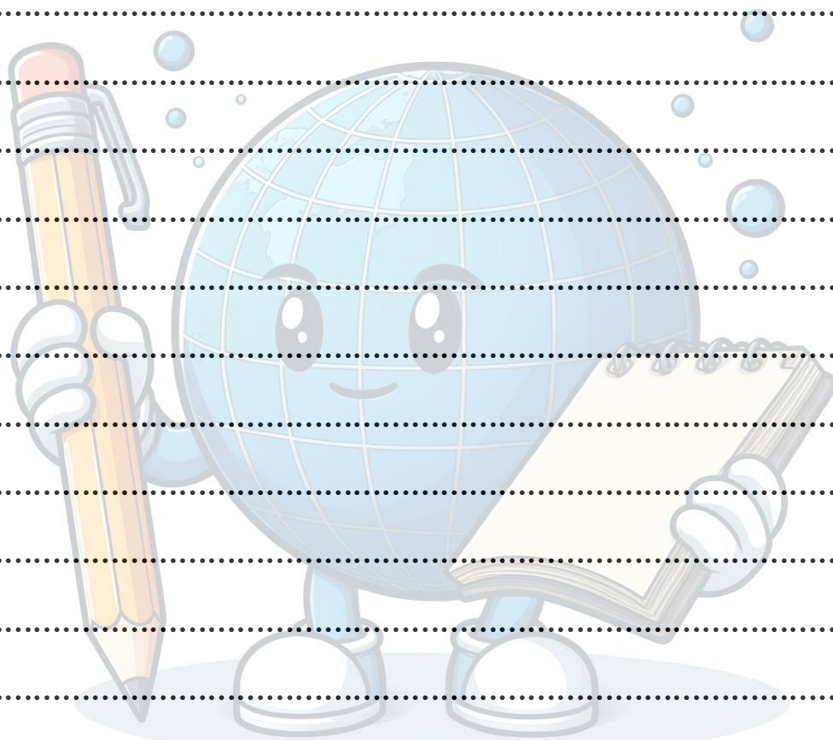
2. Hide communication from unknown persons
4. This should be made strong and changed frequently
5. This fraud inform that you have won a big sum of money
6. This type of password can be easily cracked.

⬇ Down

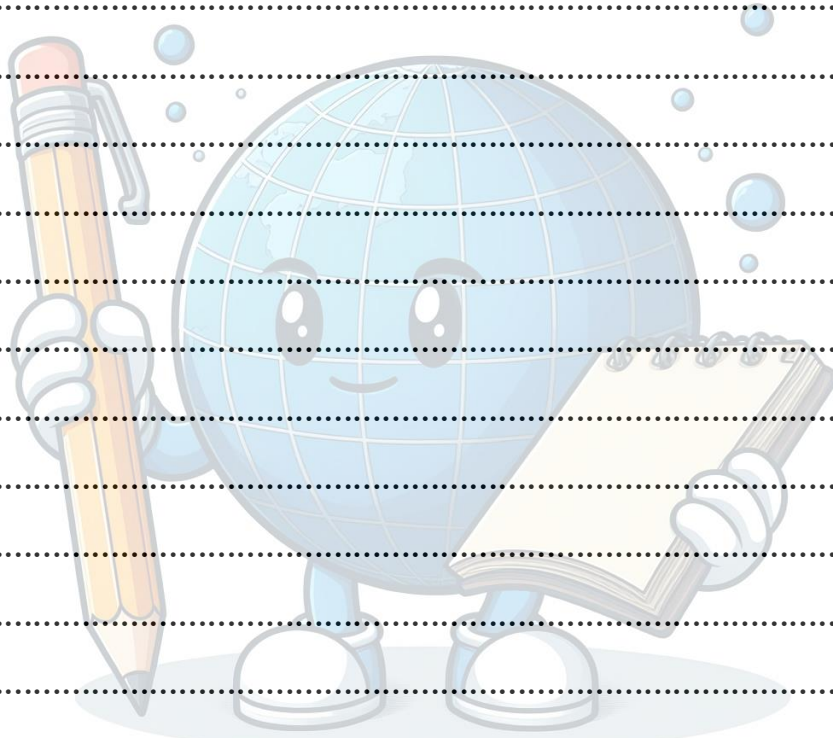
1. We all leave _____ footprint during online activities
3. Steal personal information through fraud mails
4. Never share your _____ information during online activity



Space for Notes



Space for Notes



Answer Key



Let's Practice

I. Multiple choice questions

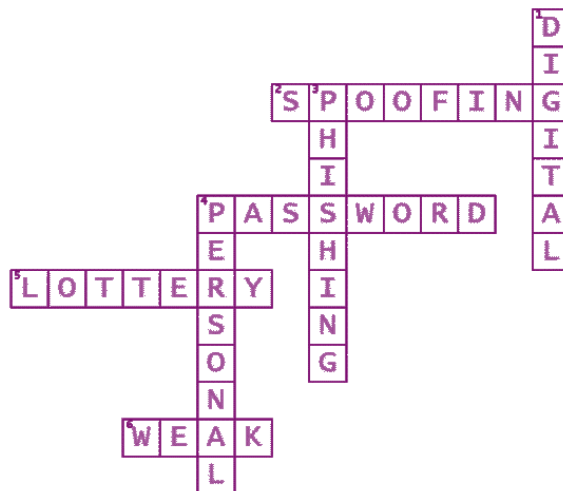
- | | | | |
|-------|-------|-------|-------|
| 1. d) | 2. a) | 3. b) | 4. d) |
| 5. b) | 6. b) | | |

II. Read through each of the following situations

- | | | | |
|--------------|-------------|--------------|--------------|
| 1. Good idea | 2. Bad idea | 3. Good idea | 4. Good idea |
| 5. Bad idea | 6. Bad idea | 7. Good idea | |

Let's Do Hands on

I. Crossword puzzle



GAMING SCAMS

1

Rohit, Just look at the prize money! This game is amazing!



2

Wow, Ritik, it's so exciting. If we win, we are going to get a huge amount of money.



3
Ya, see, I just spun the wheel, and it says I can win 50,000



4

That's wonderful! Spin again, maybe this time the reward will be double !.

5

Look Ritik, it says I have won 50,000! I just can't believe this!







Abbreviations

ALU	Arithmetic Logic Unit
CD	Compact Disk
CPU	Central Processing Unit
dpi	Dots per inch
DVD	Digital versatile disc
LAN	Local Area Network
MAN	Metropolitan Area Network
MICR	Magnetic Ink Character Recognition
MNC	Multi National Company
OCR	Optical character recognition
OS	Operating system
PAN	Personal Area Network
PC	Personal Computer
ppm	Pages per minute
RAM	Random access memory
ROM	Read only memory
SSD	Solid State Drive
TV	Television
UPS	Uninterruptible Power Supply
USB	Universal Serial Bus
VDU	Visual Display Unit
WAN	Wide Area Network
WWW	World Wide Web



CYBER FRAUDS



Online Shopping



Online Games



Weired Links



Fraud Messages



Money Scam

INFORM
**PARENTS/ ELDERS/
TEACHERS**



9531856083

The 24-hour cyber-crime prevention helpline number for the Andaman and Nicobar Islands.



1930

HELPLINE NUMBER

Report the complaint online at the National Cyber Crime Reporting Portal (NCRP)



- You can file a complaint online at cybercrime.gov.in.
- You can also report cybercrime to your [nearest police station](#).

Digital Initiatives by Govt. of India



Digital India

Power to Empower

The Digital India programme is an initiative launched by the Government of India in 2015 to make the country more digitally empowered. Its goal is to improve online services, increase internet access, and promote digital literacy among citizens.



DigiLocker

Your documents anytime, anywhere

DigiLocker is a digital platform launched by the Government of India in 2015 as part of the Digital India initiative. It allows citizens to store, access, and share important documents and certificates online in a safe and secure way.



BHIM

Bharat Interface for Money

BHIM is a digital payment app launched by the Government of India in 2016 to promote cashless transactions. It is based on the Unified Payments Interface (UPI) and allows users to send and receive money instantly using their mobile phones.



State Institute of Education
Shiksha Sadan, Sri Vijaya Puram
Andaman & Nicobar Islands

in collaboration with
State Project Office, Samagra Shiksha
UT Mission Authority, A & N Islands

